



A novel navigation system using driving state classification algorithm for semi-autonomous tractors

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ABSTRACT

The rapidly growing global population, migration, and urbanization are posing threats to the sustainability of agriculture and food security. Consequently, an autonomous driving technology for the agricultural machinery is gaining significant attention as a potential solution to these challenges. Semi-autonomous technologies such as auto-steering systems are being researched and developed. The auto-steering system has relatively low initial installation costs while still enables a tractor to automatically and precisely follow a predetermined path. However, the existing navigation systems for auto-steering system have limitations, such as constraints on initial motion for navigation system alignment, challenges in driving state identification, and excessive reliance on GPS. This paper proposes a novel navigation system integrated with a LSTM-based classification algorithm and adaptive Kalman filter, which can be applied to semi-autonomous tractors using auto-steering systems. The LSTM-based classification algorithm uses IMU data to predict whether the machinery is in stationary state, backward, or forward movements. The proposed navigation system allows the semi-autonomous tractor to perform the initial alignment during linear forward and backward motion and to apply zero velocity update effectively when the GPS signal is weak or even unavailable. The proposed algorithms were validated through extensive experiments using a tractor on both roads and farm fields. The experimental results showed that the proposed classification algorithm outperformed conventional zero velocity detectors and other machine learning methods in terms of robustness, accuracy, and versatility. The proposed navigation system experimentally demonstrated its practical applicability.

1. Introduction

The global population is expected to reach 10 billion by 2050 [1], and this population growth is expected to pose a threat to global food security [2]. Additionally, migration and urbanization threaten the sustainability of agriculture and food security [3]. Autonomous system technology in the agricultural sector is increasingly in demand as it has the potential to address these problems [3,4]. For example, researchers have shown that using autonomous agricultural machinery can result in a 24% increase in net profits compared to conventional manned machinery when both input savings and yield improvements are considered [5]. In recent years, autonomous systems for agricultural machinery have been actively researched and applied in various fields such as cultivation, sowing, harvesting, and the application of fertilizers and

pesticides [6–10]. These systems have proven their performance and efficiency, and are now being implemented in practical settings. In response to this growing demand, major agricultural machinery manufacturers are introducing autonomous systems, and in fact, more than 35,000 autonomous tractors had been deployed worldwide by 2024 [11].

Fully autonomous tractors are essentially designed to operate independently and perform tasks, which can achieve high levels of accuracy and significantly reduce labor costs. However, such systems require complex integration of multiple sensors, high performance computers, and actuators, with initial investment costs ranging from tens of thousands to hundreds of thousands of dollars [12–14]. These significant upfront costs can be barriers for small and medium-sized farms, which can make the usage of fully autonomous tractors difficult. On the other hand, semi-autonomous tractors with an auto steering system offer a

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Nomenclature			
w_e	Earth's rotational angular velocity (rad/s)	P	Error covariance matrix of state error
M	Radius of curvature in median (m)	Q	Process noise matrix
N	Radius of curvature in prime vertical (m)	R	Measurement noise matrix
φ	Latitude (rad)	H	Measurement matrix
λ	Longitude (rad)	z	Measurement vector
h	Altitude (m)	K	Kalman gain
v_N	Northward velocity (m/s)	v	Innovation term
v_E	Eastward velocity (m/s)	x	Input vector of LSTM layer
v_D	Downward velocity (m/s)	S	Driving state
C_b^n	Direction cosine matrix from body frame to navigation frame	G	Set of driving states
f	Specific force vector (m/s^2)	Y	Maneuver state
w	Angular rate vector (rad/s)	U	Set of maneuver states
Ω	Skew symmetric matrix of $w(rad/s)$	J	Weight matrix of hidden state
g	Magnitude of gravity (m/s^2)	W	Weight matrix of input vector
n	Navigation frame	B	Bias vector of LSTM layer
b	Body frame	σ	Sigmoid function
e	Earth frame	$\tanh$	Hyperbolic tangent function
i	Inertial frame	s	Softmax activation function
Δt	Sampling time (s)	$\tilde{C}$	Cell state candidate
r	Position vector (rad, m)	C	Cell state
v	Velocity vector (m/s)	p	Output of input gate
δx	State error vector	O	Output of output gate
δr	Position error vector (rad, m)	E	Output of forget gate
δv	Velocity error vector (m/s)	m	Hidden state of LSTM layer
ϵ	Attitude error vector (rad)	L	Length of sequence data
b^a	Bias vector of acceleration (m/s^2)	y^a	Accelerometer output
b^w	Bias vector of gyroscope (rad/s)	y^w	Gyroscope output
F	Error system matrix in continuous time	T	Function value of zero velocity detector
A	Error system matrix in discrete time	σ_a	Noise standard deviation of accelerometer (m/s^2)
		σ_w	Noise standard deviation of gyroscope (rad/s)
		γ	Threshold of zero velocity detector

cost-effective alternative. Although it still requires some operator intervention, the auto-steering system allows tractors to automatically and accurately follow preset routes, ensuring both economic and functional performance while significantly reducing operator workload.

As shown in Fig. 1, the auto-steering system for tractors can be implemented simply by replacing the existing steering wheel with an

electric steering wheel that integrates the controller and sensor [15]. Therefore, the installation costs are relatively low, ranging from thousands to tens of thousands of dollars [16,17]. The system also offers a variety of benefits, including reduced fuel consumption and labor requirements, increased profitability, reduced costs, and improved operational efficiency [18–20]. Auto-steering systems for semi-autonomous

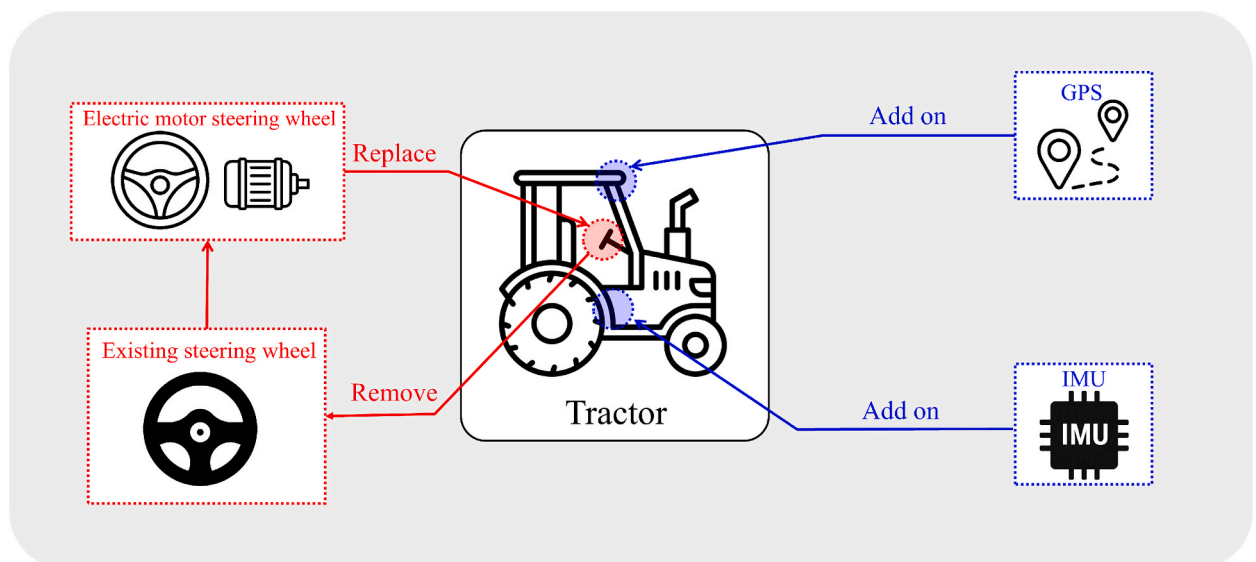


Fig. 1. Auto-steering system for tractors with electric steering wheel replacement.

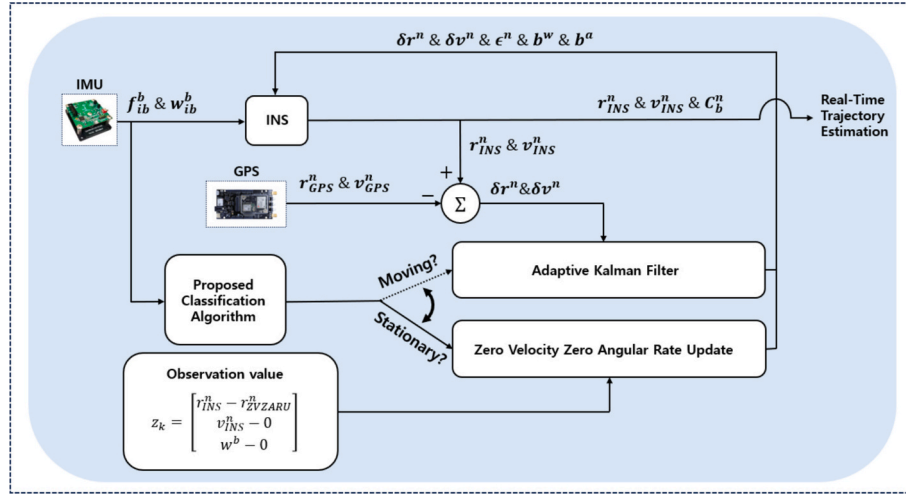


Fig. 2. Overview of navigation system using proposed classification algorithm.

tractors rely on navigation systems based on Global Positioning System (GPS) and Inertial Measurement Unit (IMU) to automatically control the steering along a predetermined path. In order to enhance the accuracy of the navigation system, a real-time kinematic GPS (RTK-GPS) has been utilized [21]. Researchers showed that the auto-steering system using the RTK-GPS and IMU provides centimeter-level accuracy and thus reduces overlap and gaps during operations [21]. However, the existing navigation systems for auto-steering system have limitations, such as constraints on initial motion for navigation system alignment, challenges in driving state identification, and excessive reliance on GPS.

In the navigation system of the semi-autonomous tractor, determining the driving states such as stationary state, forward, and backward movements is crucial. Identifying when the tractor is stationary allows for the application of correction algorithms like Zero Velocity Update (ZUPT). ZUPT restrains the divergence of estimation error in navigation algorithms and ensures precise position and orientation estimation [22]. Additionally, recognizing forward and backward motions is necessary for properly conducting the initial alignment of the navigation system. Fully autonomous tractors can determine their driving state by utilizing GPS alongside in-vehicle data about wheel speed, gear, transmission, and braking. However, semi-autonomous tractors equipped with auto-steering systems cannot access in-vehicle data and currently rely only on GPS to identify driving states. As a result, most GPS-based commercial auto-steering systems available on the market can only perform initial alignment of the navigation system during forward motion. In addition, the low update frequency of GPS measurements prevents the application of ZUPT between consecutive GPS updates. Furthermore, the system that relies solely on GPS may lead to problems if the GPS signal is weak or lost. Therefore, the ability to determine driving states including stationary state, forward, and backward movements without requiring GPS data would enhance the robustness of the navigation system by enabling the semi-autonomous tractor to perform the initial alignment during any motion and to apply ZUPT effectively even when the GPS signal is weak or unavailable.

Motivated by these limitations, researchers have considered the use of IMUs as an alternative sensing modality for identifying vehicle motion states to improve the robustness of navigation performance. However, studies that explore the use of IMUs to determine the stationary and moving states of vehicles are still limited [23,24]. Most studies have focused on detecting the stationary and moving states of pedestrians using IMUs attached to the foot. Zero Velocity Detector (ZVD) for distinguishing between a pedestrian's moving and stationary state has been studied using various approaches, employing artificial intelligence techniques, or utilizing probabilistic methods [25–29]. These methods, however, are not designed to differentiate between forward and

backward motions. Most importantly, they are challenging to use for determining whether agricultural machinery is stationary due to severe vibrations and the distinct driving characteristics of agricultural machinery compared to those of pedestrians and other road vehicles.

In this paper, we propose a novel navigation system integrated with a Long Short-Term Memory (LSTM)-based classification algorithm for semi-autonomous tractors using the auto-steering system. An LSTM-based classification algorithm integrated with a finite state machine (FSM) is developed to determine the driving states of a semi-autonomous tractor. The algorithm uses only IMU data to identify whether the machinery is currently in a stationary state, forward, or backward movements in real-time. By integrating the proposed classification algorithm with the adaptive Kalman filter-based navigation framework, we enable initial alignment not only during forward motion but also during backward motion, which is not possible in conventional GPS-dependent approaches. Moreover, this approach allows for more accurate stationary state determination compared to conventional ZVDs and provides robust navigation performance across various driving surface conditions. Experimental datasets were collected from tractor operations on both roads and farm fields, and the performance of the proposed algorithm was extensively evaluated through comparisons with conventional ZVDs and other machine learning-based methods. Finally, the performance of position and attitude estimation by the proposed navigation system is experimentally validated using real tractor operations on roads and farm fields. The contribution of this paper is as follows:

- 1) This paper proposes an IMU-only LSTM-FSM driving state classifier that identifies stationary, forward, and backward motions without reliance on GPS, and validates its effectiveness through real-world experiments.
- 2) This paper integrates the proposed classifier into an adaptive Kalman filter framework, which allows initial alignment to be performed during both forward and backward motion.
- 3) This paper improves the robustness of motion and stationary state determination by ensuring that ZUPT can be reliably applied even when GPS is degraded or unavailable.

The structure of this paper is as follows. Section 2 introduces the navigation algorithms used in this study, such as the adaptive Kalman filter and Zero Velocity Zero Angular Rate Update (ZVZARU). Section 3 proposes an LSTM-based classification algorithm for classifying the driving states of the agricultural machinery. Section 4 presents the results of the proposed algorithm, trained on data obtained from experiments using actual tractor operations, through comparisons with

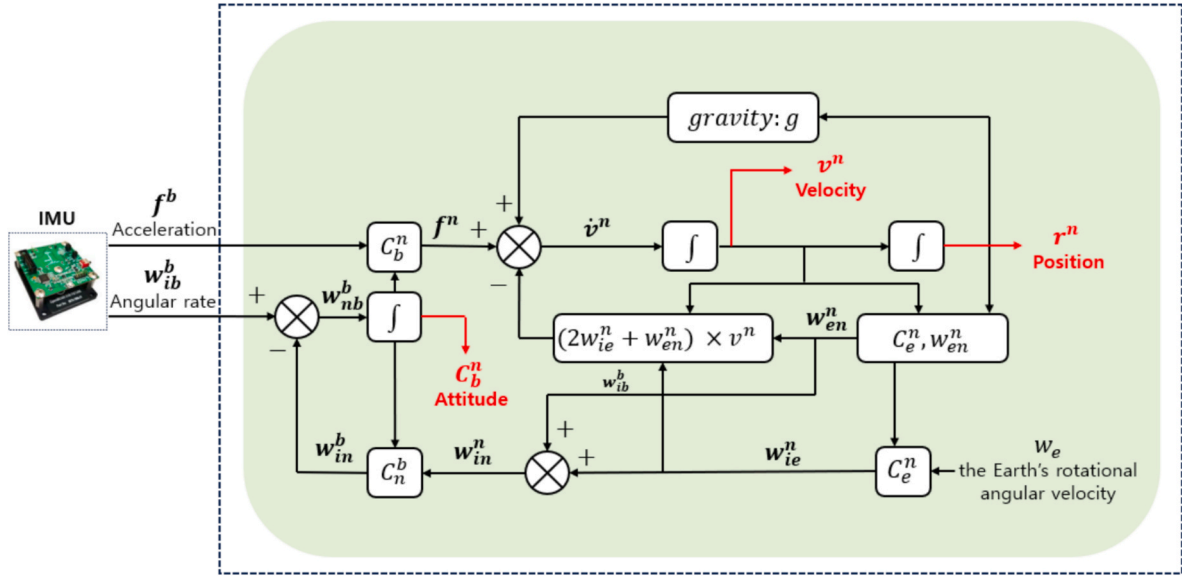


Fig. 3. INS mechanization of navigation frame.

conventional ZVDs and other machine-learning based methods. The results of the position and attitude estimation using the navigation system integrated with the proposed algorithm are also presented. Finally, Section 5 discusses the conclusion and future work.

2. INS/GPS integrated navigation algorithm using adaptive Kalman filter

Inertial Navigation System (INS) uses an IMU to calculate position and attitude at a high frequency using gyroscopes and accelerometers, without being constrained by external signals or locations. However, biases and errors in the IMU accumulate during the integration process, which poses a risk of increasing error over long-term usage. On the other hand, GPS has the advantage of measuring position without error accumulation over long periods, but it is heavily constrained by external signals and locations and measures at a low frequency. Therefore, integrating these two sensors can complement each other's strengths and weaknesses. In this paper, a loosely-coupled integrated navigation system is utilized for the integration of INS and GPS. When the agricultural machinery is moving, the adaptive Kalman filter is used, and when the machinery is stationary, Zero Velocity Zero Angular Rate Update (ZVZARU) is applied. To apply ZVZARU, it is necessary to determine whether the machinery is stationary or moving. The classification algorithm proposed in Section 3 addresses this issue. Fig. 2 shows the overview of the navigation algorithm integrated with the proposed classification algorithm.

2.1. Inertial navigation system

INS calculates a position, velocity, and attitude by integrating the measurements from accelerometers and gyroscopes, as shown in Fig. 3. The differential equations of INS are as follows [30]:

$$\dot{r}^n = \begin{bmatrix} \dot{\varphi} \\ \dot{\lambda} \\ \dot{h} \end{bmatrix} = \begin{bmatrix} \frac{1}{M+h} & 0 & 0 \\ 0 & \frac{1}{(N+h)\cos\varphi} & 0 \\ 0 & 0 & -1 \end{bmatrix} \begin{bmatrix} v_N \\ v_E \\ v_D \end{bmatrix} \quad (1)$$

$$\dot{v}^n = C_b^n f^b - (2w_{ie}^n + w_{en}^n) \times v^n + g^n \quad (2)$$

$$\dot{C}_b^n = C_b^n (\Omega_{ib}^b - \Omega_{in}^b) \quad (3)$$

with

$$w_{ie}^n = [w_e \cos\varphi \quad 0 \quad -w_e \sin\varphi]^T \quad (4)$$

$$w_{en}^n = [v_E/(N+h) - v_N/(M+h) - v_E \tan\varphi/(N+h)]^T \quad (5)$$

$$w_{in}^n = w_{ie}^n + w_{en}^n \quad (6)$$

where $r^n = [\varphi \quad \lambda \quad h]^T$ denotes the position in the n-frame, representing latitude, longitude, and altitude, respectively, $v^n = [v_N \quad v_E \quad v_D]^T$ represents the velocity in the n-frame, corresponding to the northward, eastward, and downward velocities, w_e represents the earth's rotational angular velocity, C_b^n denotes the direction cosine matrix from the body frame to the navigation frame and Ω represents the skew symmetric matrix of w .

2.2. Adaptive Kalman filter

In this paper, errors of the INS and the sensor biases of the accelerometer and gyroscope are estimated and used to enhance the navigation performance. The state vector of the errors is defined as

$$\delta x = [\delta r^n \quad \delta v^n \quad \epsilon^n \quad b^w \quad b^a]^T \quad (7)$$

where r^n , v^n , ϵ^n , b^w , and b^a are the position error, velocity error, attitude error, gyroscope bias, and accelerometer bias, respectively. The INS error equations are as follows [30]:

$$\delta \dot{r}^n = F_r \delta r^n + F_v \delta v^n \quad (8)$$

$$\delta \dot{v}^n = v^n \times (2\delta w_{ie}^n + \delta w_{en}^n) + \delta g^n - (2w_{ie}^n + w_{en}^n) \times \delta v^n + f^n \times \epsilon^n + C_b^n \delta f^b \quad (9)$$

$$\dot{\epsilon}^n = \delta w_{in}^n - (w_{in}^n \times) \epsilon^n - C_b^n \delta w_{ib}^b \quad (10)$$

with

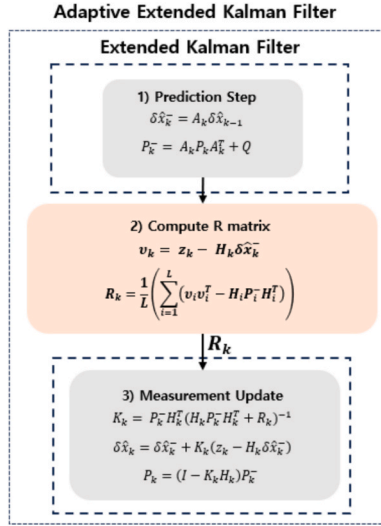


Fig. 4. Structure of adaptive Kalman filter.

$$F_{rr} = \begin{bmatrix} 0 & 0 & \frac{-v_N}{(M+h)^2} \\ \frac{v_E \sin \varphi}{(N+h) \cos^2 \varphi} & 0 & \frac{-v_E}{(N+h)^2 \cos \varphi} \\ 0 & 0 & 0 \end{bmatrix} \cdot F_{rv}$$

$$= \begin{bmatrix} \frac{1}{M+h} & 0 & 0 \\ 0 & \frac{1}{(N+h) \cos \varphi} & 0 \\ 0 & 0 & -1 \end{bmatrix} \quad (11)$$

Based on the state vector and error equations, an adaptive Kalman filter is employed when the agricultural machinery is moving. When the agricultural machinery is stationary, the ZVZARU is used, leveraging the fact that both the velocity and angular velocity are zero, and using this as the measurement value [31].

In the case of a standard Kalman filter, the R matrix, which represents the measurement noise covariance, is assumed to be constant. However, in the case of agricultural machinery, the road conditions in fields and paddies are highly irregular, and the performance of GPS can degrade when the signal is interrupted. Therefore, an adaptive Kalman filter, which adaptively estimates the R matrix considering sensors performance, can achieve better estimation performance in such uncertain environments [32–34]. The adaptive Kalman filter used in this paper consists of the prediction step, measurement update, and R matrix computation step. The prediction step involves predicting the current state error and error covariance based on the previous state error and system matrix, and is formulated by the following equations:

$$\delta\hat{x}_k^- = A_k \delta\hat{x}_{k-1} \quad (12)$$

$$P_k^- = A_k P_{k-1} A_k^T + Q \quad (13)$$

where, A represents the system matrix obtained by linearization of the INS error equations in discrete time, and Δt denotes the sampling time. Additionally, P and Q refer to the error covariance matrix and process noise covariance matrix, respectively. The subscript k indicates the current time step. The measurement update step involves updating the predicted state error using the measured values and is formulated by the following equations:

$$K_k = P_k^- H_k^T (H_k P_k^- H_k^T + R_k)^{-1} \quad (14)$$

$$z_k = \begin{bmatrix} r_{INS}^n - r_{GPS}^n \\ v_{INS}^n - v_{GPS}^n \end{bmatrix} = \begin{bmatrix} \varphi_{INS} - \varphi_{GPS} \\ \lambda_{INS} - \lambda_{GPS} \\ h_{INS} - h_{GPS} \\ v_{N,INS} - v_{N,GPS} \\ v_{E,INS} - v_{E,GPS} \\ v_{D,INS} - v_{D,GPS} \end{bmatrix} \quad (15)$$

$$\delta\hat{x}_k = \delta\hat{x}_k^- + K_k (z_k - H_k \delta\hat{x}_k^-) \quad (16)$$

$$P_k = (I_{15} - K_k H_k) P_k^- \quad (17)$$

where K represents the Kalman gain, and R and H refer to the measurement noise covariance matrix and measurement matrix, respectively. Additionally, z denotes the difference between the value calculated by the INS mechanization and the measurement from the GPS, with the subscripts *INS* and *GPS* indicating the values from the INS mechanization and the GPS sensor measurement, respectively. I_{15} represents a 15×15 identity matrix. Before the measurement update, R is calculated based on the innovation term:

$$v_k = z_k - H_k \delta\hat{x}_k^- \quad (18)$$

$$R_k = \frac{1}{L} \left(\sum_{i=1}^L (v_i v_i^T - H_i P_i^- H_i^T) \right) \quad (19)$$

where v_k represents the innovation term based on measurement values, and L refers to the length of sequence data for calculating the R matrix. Fig. 4 illustrates the structure of this adaptive Kalman filter.

2.3. Zero velocity zero angular rate update

When the agricultural machinery is stationary, the GPS velocity should theoretically be zero. However, in practice, these measurements include noise and the GPS measures a non-zero velocity even when the machinery is stationary. These values can lead to error accumulation, making state estimation inaccurate. Therefore, in this paper, we implement ZVZARU using three measurements when the machinery is stationary. Firstly, for the velocity, we use a value of zero instead of the GPS velocity. Secondly, since the gyroscope's angular rate can be assumed to be the bias of the gyroscope when the agricultural machinery is stationary, this value is used as the measurement for the gyroscope's bias. Lastly, the estimated position just before the machinery stopped is used as the position measurement. ZVZARU consists of two steps: the prediction step and the measurement step. The equations for the prediction step are given in (12) and (13), and the equations for the measurement step are given in (14), (16), and (17). The measurement vector are as follows:

$$z_k = \begin{bmatrix} \varphi_{INS} - \varphi_{ZVZARU} \\ \lambda_{INS} - \lambda_{ZVZARU} \\ h_{INS} - h_{ZVZARU} \\ v_{N,INS} - 0 \\ v_{E,INS} - 0 \\ v_{D,INS} - 0 \\ w_x^b - 0 \\ w_y^b - 0 \\ w_z^b - 0 \end{bmatrix} = \begin{bmatrix} r_{INS}^n - r_{ZVZARU}^n \\ v_{INS}^n - 0 \\ w^b - 0 \end{bmatrix} \quad (20)$$

where r_{ZVZARU}^n represents the position obtained from just before stopping, and w^b refers to the gyroscope measurement.

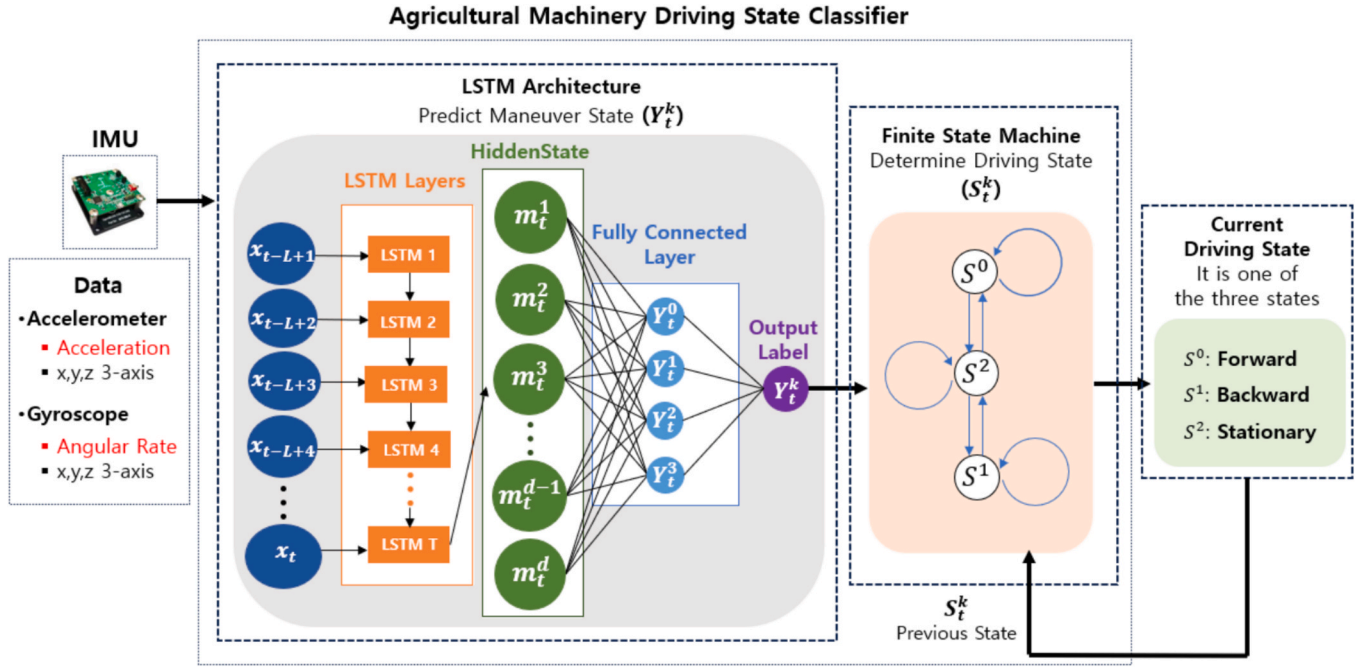


Fig. 5. Architecture of proposed classification algorithm.

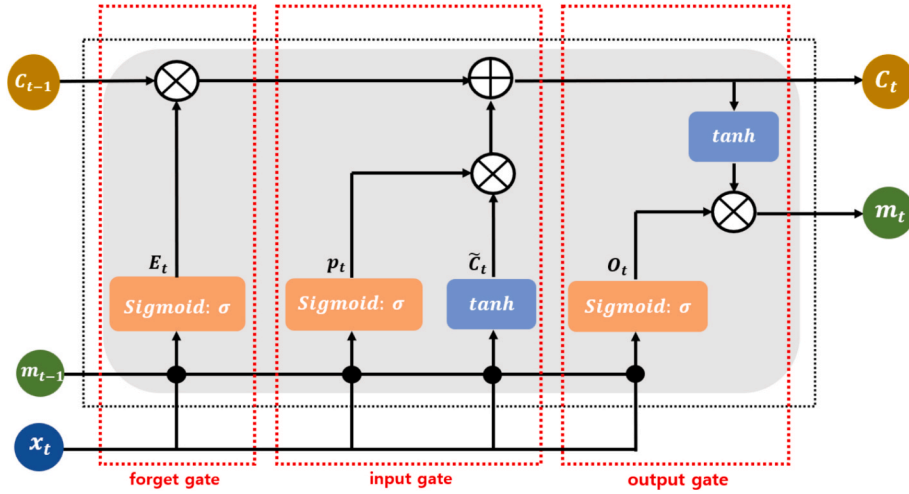


Fig. 6. Structure of LSTM cell.

3. Driving state classification algorithm for agricultural machinery

This paper presents a Long Short Term Memory (LSTM)-based algorithm for classifying the driving states of an agricultural machinery. The algorithm is composed of a combination of LSTM, a type of artificial neural network, and a Finite State Machine (FSM) that determines the state based on conditions, as shown in Fig. 5. The LSTM receives 3-axis acceleration data and 3-axis angular rate data from the IMU as input. The output of the LSTM is the current maneuver state of the agricultural machinery (Startgo, Startback, Move, or Stop). This output then serves as input to the FSM, which, based on predefined rules, ultimately determines the current driving state of the agricultural machinery (Forward, Backward, or Stationary). In this paper, we use 'Stop' and 'Stationary' to denote distinct output classes of the LSTM and FSM, respectively. While both terms indicate that agricultural machinery is not moving, 'Stop' specifically refers to an intermediate decision made by the LSTM model, whereas 'Stationary' represents the final decision

from the proposed classification algorithm.

3.1. Maneuver prediction using LSTM

LSTM is a type of recurrent neural network (RNN). While RNNs face the issue of vanishing gradients when handling long sequence data, making it difficult to manage long-term dependencies, LSTM overcomes this by having memory cells that can remember information over long periods [35]. This enables LSTMs to stably learn long-term dependencies and efficiently handle long time-series data. The effectiveness of LSTMs has been demonstrated in various dynamic system applications, including dynamic response prediction, complex time-series forecasting, and fault diagnosis [36–38]. Therefore, these characteristics make LSTMs particularly suitable for predicting the maneuvering states of agricultural machinery, where the current state is closely related to past states.

A LSTM layer consists of multiple memory cells, each with a forget gate, an input gate, and an output gate as shown in Fig. 6. In the forget

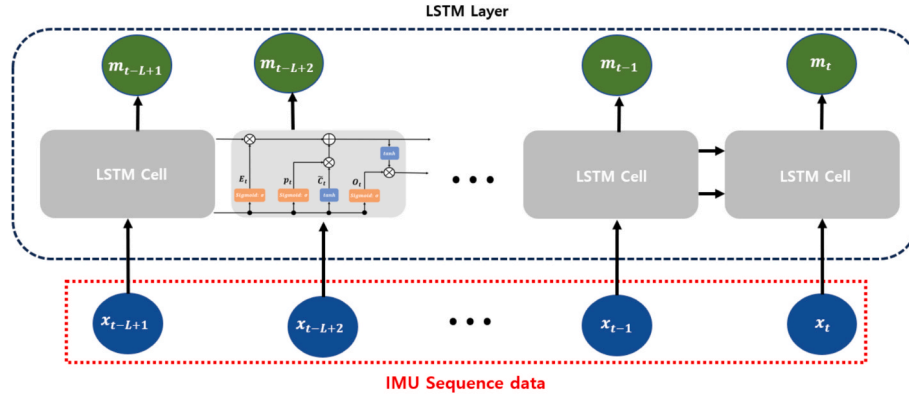


Fig. 7. Structure of LSTM layer.

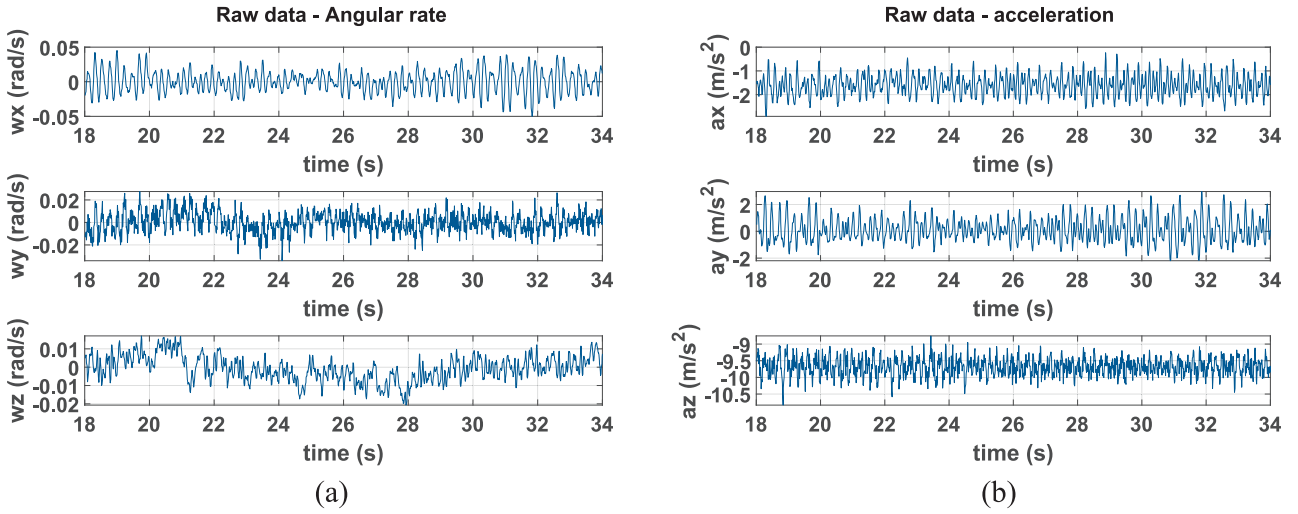


Fig. 8. (a) Raw angular rate and (b) acceleration data from IMU on a tractor while it is moving backward.

gate, an unnecessary information from the past is discarded, while a necessary information is retained. The equation for the forget gate is as follows:

$$E_t = \sigma(W_E x_t + J_E m_{t-1} + B_E) \quad (21)$$

where m is the hidden state and $x = [y_x^a \ y_y^a \ y_z^a \ y_x^w \ y_y^w \ y_z^w]^T$ represents the input vector. The input vector refers to the measurements from the accelerometer and gyroscope. y_x^a , y_y^a , and y_z^a represent the accelerometer measurements along the x , y , and z axis, respectively, while y_x^w , y_y^w , and y_z^w represent the gyroscope measurements along the x , y , and z axis, respectively. W_E and J_E are weight matrices related to the input vector and the hidden state in the forget gate, respectively. Also, B_E is a bias vector in the forget gate. E indicates the output of the forget gate, also t and $t-1$ represents the current time step and the previous time step, respectively. The input gate determines how much of the new information at the current time step should be remembered and updates the cell state. The equations for the input gate are as follows:

$$\tilde{C}_t = \tanh(W_C x_t + J_C m_{t-1} + B_C) \quad (22)$$

$$p_t = \sigma(W_p x_t + J_p m_{t-1} + B_p) \quad (23)$$

$$C_t = E_t \odot C_{t-1} + p_t \odot \tilde{C}_t \quad (24)$$

where $\tilde{C}$ and C represent the cell state candidate and the cell state, respectively. p indicates the output of the input gate. W_C and J_C are

weight matrices related to the input vector and the hidden state in the cell state, respectively. Also, B_C is a bias vector in the cell state. W_p and J_p are weight matrices related to the input vector and the hidden state in the input gate, respectively. Also, B_p is a bias vector in the input gate. $\odot$ is point-wise multiplication. The output gate is responsible for determining how much information should be sent out as output. The equations for the output gate are as follows:

$$O_t = \sigma(W_O x_t + J_O m_{t-1} + B_O) \quad (25)$$

$$m_t = O_t \odot \tanh(C_t) \quad (26)$$

where O indicates the output of the output gate. W_O and J_O are weight matrices related to the input vector and the hidden state in the output gate, respectively. Also, B_O is a bias vector in the output gate.

An LSTM layer is composed of multiple LSTM cells, each of which processes a single time step of the sequence data and passes the information to the next LSTM cell. Fig. 7 shows the structure of such an LSTM layer. In Fig. 7, L represents the length of the sequence data.

The hidden states m extracted from the LSTM cells are used as inputs to the subsequent fully connected layer. After that, the output of the fully connected layer is then passed through a softmax activation function, which converts the logits into probabilities for each class (Stop, Startback, Startgo, and Move), and the class with the highest probability is selected as the final prediction Y^i . The softmax activation function is defined as

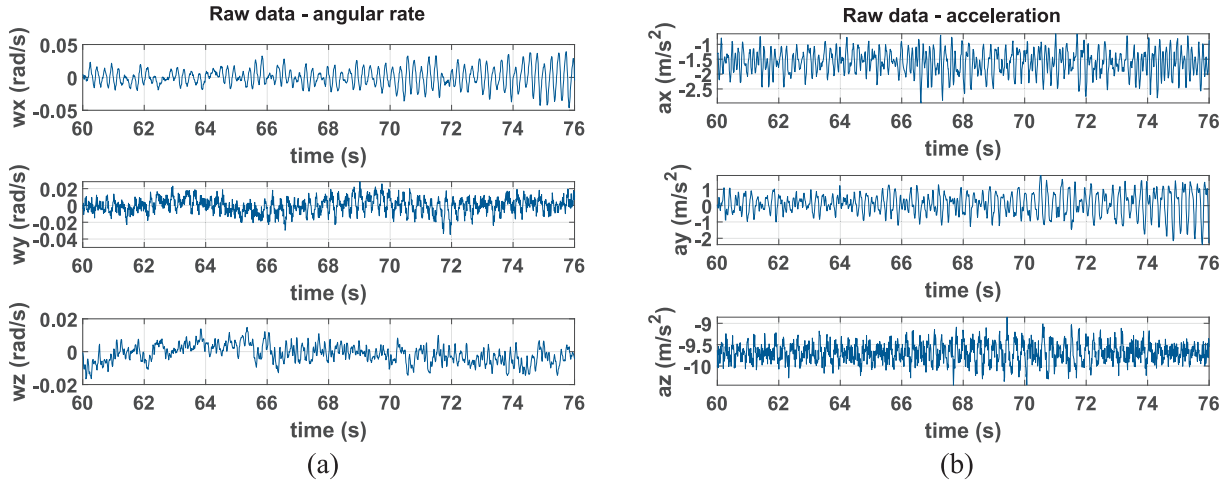


Fig. 9. (a) Raw angular rate and (b) acceleration data from IMU on a tractor while it is moving forward.

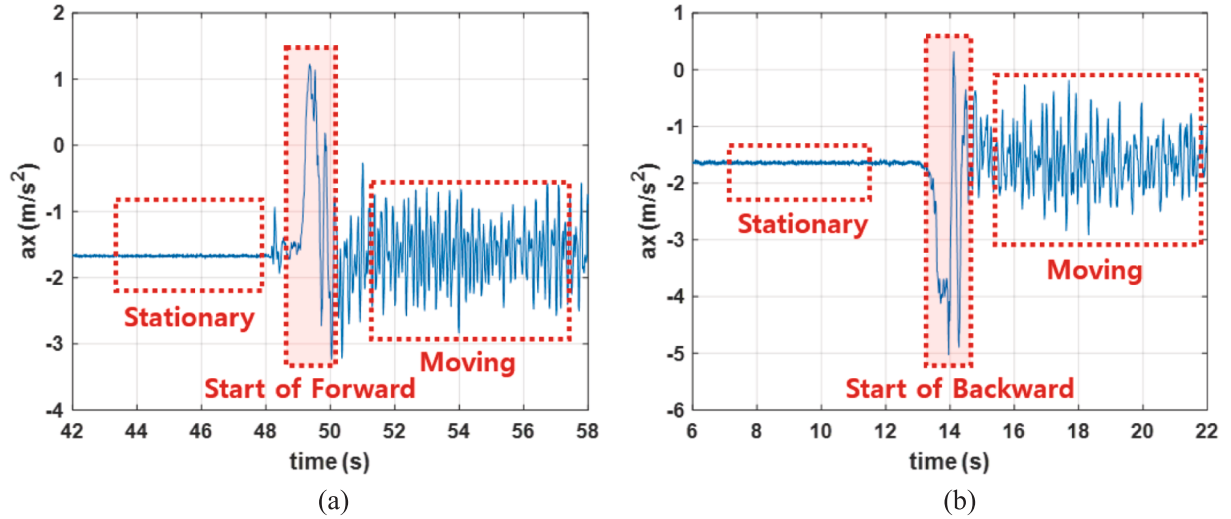


Fig. 10. The measurement values of the x-axis accelerometer when (a) moving forward and (b) moving backward.

$$s(Y^i) = \frac{e^{Y^i}}{\sum_{j=1}^n e^{Y^j}}, i = 0, 1, 2, 3 \quad (27)$$

where i is the index for each class: Stop, Startback, Startgo, and Move. The output value of the softmax function for each class ranges between 0 and 1, and the sum of these values is 1.

Here, Startgo and Startback represent the start of forward driving and backward driving, respectively. As shown in Figs. 8 and 9, distinguishing directly between forward driving and backward driving is difficult using only the accelerometer and gyroscope data from the IMU.

If the agricultural machinery is driving in a constant velocity without accelerating or decelerating, it becomes even more difficult to distinguish in real-time whether it is moving forward or backward. However, when the agricultural machine starts moving backward from a stop or starts moving forward from a stop, as shown in Fig. 10, the difference between the two cases can be distinguished from the acceleration of the agricultural machinery along x-axis. When the agricultural machinery starts moving forward from a stop, the measurement of the accelerometer rises significantly and then returns as shown in Fig. 10(a). In contrast, when the machinery starts moving backward from a stop, the measurement of the accelerometer along x-axis drops significantly and then returns as shown in Fig. 10(b).

Based on these observations, Startgo and Startback are utilized as the

output class of the LSTM. As a result, the LSTM model predicts the moment at which the machine starts moving backward or moving forward. These predicted values are then used as input to the FSM, which ultimately determines whether the agricultural machinery is currently moving backward or forward.

3.2. Integrating LSTM with FSM for driving state determination

The FSM is a mathematical model used to design systems with a finite number of states and has been employed in algorithms for mobility applications [39,40]. It transitions from a state at the previous time step $t-1$ to the current state at time t upon receiving a specific input. In this paper, we use a Deterministic Finite State Machine (DFSM), which transitions to a determined next state for a specific input in a specific state. The previous state represents the driving state of the agricultural machinery at the previous time step, and it transitions to the driving state at the current time step based on the given input. The previous state is defined by the following set:

$$S_{t-1}^k \in G_{t-1} = \{S_{t-1}^0, S_{t-1}^1, S_{t-1}^2\} \quad (28)$$

where G_{t-1} represents the set of driving states at the previous time step, and S_{t-1}^0 , S_{t-1}^1 , and S_{t-1}^2 represent the FSM output at the previous time

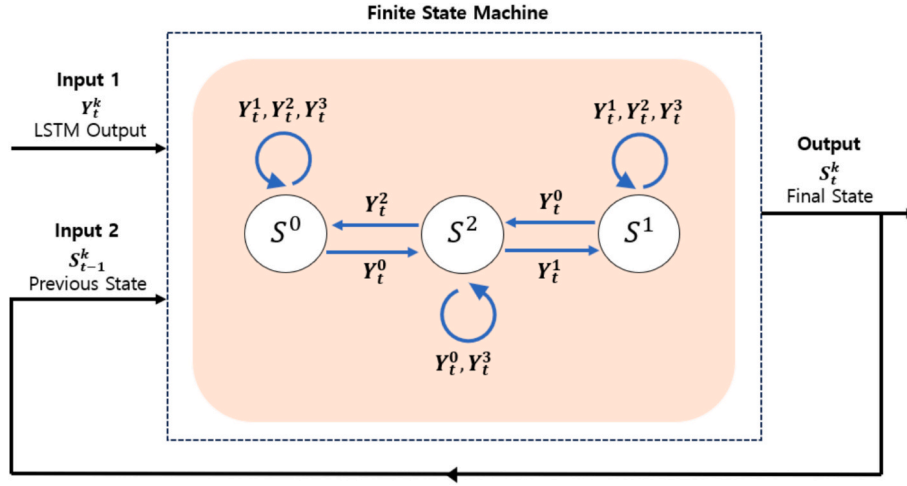


Fig. 11. Structure of proposed finite state machine.

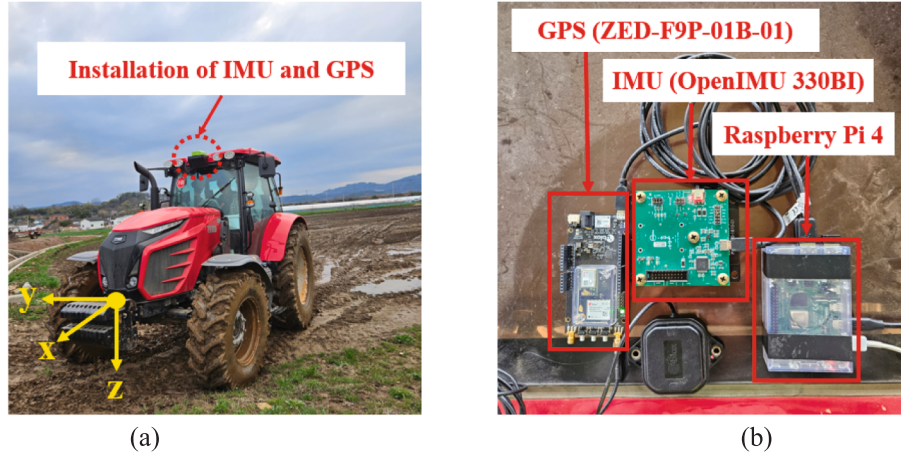


Fig. 12. (a) Tractor and (b) sensor system used in the experiment.

step: Forward, Backward, and Stationary, respectively. The input refers to the output at the current time step from the LSTM and is defined by the following set:

$$Y_t^k \in U = \{Y_t^0, Y_t^1, Y_t^2, Y_t^3\} \quad (29)$$

where U represents the set of the predicted classes of the LSTM at the current time step. Here, Y_t^0 represents Stop, Y_t^1 represents Startback, Y_t^2 represents Startgo, and Y_t^3 represents Move. The final state refers to the final output determined by the FSM and represents the current driving state of the agricultural machinery, and it is defined as follows:

$$S_t^k \in G_t = \{S_t^0, S_t^1, S_t^2\} \quad (30)$$

where S_t^0, S_t^1 , and S_t^2 represent the FSM output at the current time step: Forward, Backward, and Stationary, respectively, and G_t represents the set of driving states at the current time step. The structure of this FSM is shown in Fig. 11.

Transitions between states of the proposed FSM are as follows:

- (1) If the driving state in the previous time step is Stationary, the FSM will only determine the current driving state as Forward or Backward when it receives a Startgo or Startback signal from the LSTM. In other words, the starting point when transitioning from a stationary state to moving state is the moment when the LSTM outputs

a Startgo or Startback signal. Otherwise, the driving state will remain the same:

$$\begin{cases} S_{t-1}^2 \rightarrow S_t^1, & \text{if } Y_t^k = Y_t^1 \\ S_{t-1}^2 \rightarrow S_t^0, & \text{if } Y_t^k = Y_t^2 \\ S_{t-1}^2 \rightarrow S_t^2, & \text{otherwise} \end{cases} \quad (31)$$

- (2) If the driving state in the previous time step is a Forward and the FSM receives a Stop signal from the LSTM, the FSM will determine that the current driving state is Stationary. Otherwise, it will

Table 1

Specifications of the IMU and GPS used in the experiment [41,42].

Sensor	Parameter	Value
Accelerometer (OpenIMU 330BI)	Update rate	100Hz
	Range	$\pm 8g$
	Resolution	$< 0.5mg$
	Bias Instability	$20\mu g$
	Update rate	100Hz
Gyroscope (OpenIMU 330BI)	Range	$\pm 400^\circ/s$
	Resolution	$< 0.02^\circ/s$
	Bias Instability	$1.5^\circ/hr$
	Update rate	8Hz
GPS (U-blox ZED-F9P-01B-01)	Horizontal position accuracy	$0.01m + 1ppm$
	Vertical Position accuracy	$0.01m + 1ppm$



Fig. 13. (a) Roads and (b) farm fields used in the experiment.

Table 2
Test data information.

Test dataset		
Trajectory #	Driving location	Driving state
T1	Road	$S^2 \rightarrow S^1 \rightarrow S^2 \rightarrow S^0 \rightarrow S^2$
T2	Road	$S^2 \rightarrow S^1 \rightarrow S^2 \rightarrow S^0 \rightarrow S^2$
T3	Road & Farm Field	$S^2 \rightarrow S^1 \rightarrow S^2 \rightarrow S^0 \rightarrow S^2$

maintain the current driving state as the same as the previous time step:

$$\begin{cases} S_{t-1}^1 \rightarrow S_t^2, & \text{if } Y_t^k = Y_t^0 \\ S_{t-1}^1 \rightarrow S_t^1 & \text{otherwise} \end{cases} \quad (32)$$

3) If the driving state in the previous time step is a Backward, the state transition to Stationary occurs when the FSM receives a Stop signal from the LSTM. Otherwise, it will maintain the current driving state as the same as the previous time step:

$$\begin{cases} S_{t-1}^0 \rightarrow S_t^2, & \text{if } Y_t^k = Y_t^0 \\ S_{t-1}^0 \rightarrow S_t^0 & \text{otherwise} \end{cases} \quad (33)$$

It is assumed that the driving state of the agricultural machinery does not directly switch from Backward to Forward or vice versa. In other words, the agricultural machinery must undergo a stationary phase during the transition between forward and backward motions. This assumption is reasonable because most agricultural machinery, such as tractors, requires the machine to stop momentarily during gear changes. Also, it is noted that in order to prevent the current driving state from immediately transitioning to an incorrect driving state due to inaccurate

state predictions, we set a count for Stop, Startback, and Startgo. When the driving state transitions from Stationary to Forward or Backward, the state only changes after Startgo or Startback has appeared consecutively for a predefined number of occurrences. Similarly, when transitioning from Forward or Backward to Stationary, a count is also set for Stationary.

4. Experimental results

4.1. Experimental setup

In order to train and validate the proposed algorithms, experiments were conducted by using a tractor equipped with IMU, GPS, and microprocessor. The IMU and GPS used in the experiment are the OpenIMU330BI from Aceinna and the ZED-F9P-01B-01 from U-blox, with the GPS utilizing RTK functionality. The IMU collected data at a frequency of 100 Hz, while the GPS collected data at 8 Hz. An embedded system using a Raspberry Pi 4 was developed to process sensor data and proposed algorithms in real-time. Fig. 12 shows the tractor operated in the experiment along with the installed IMU and GPS. Table 1 provides the sensor specifications for the IMU and GPS used in the experiment. In Fig. 12(a), the yellow arrow lines indicate the IMU coordinate axes relative to the tractor body. The IMU is mounted such that its x-axis is aligned with the forward direction of the tractor.

Experimental data were collected by operating the tractor on both roads and farm fields. Fig. 13 shows the roads and farm fields where the experiments were conducted. A total of 16 trajectory were driven, out of which 13 were used as train data and the remaining 3 were used as test data (a total of 379,738 datasets were obtained and used). Table 2 presents the driving locations and states for the trajectories included in

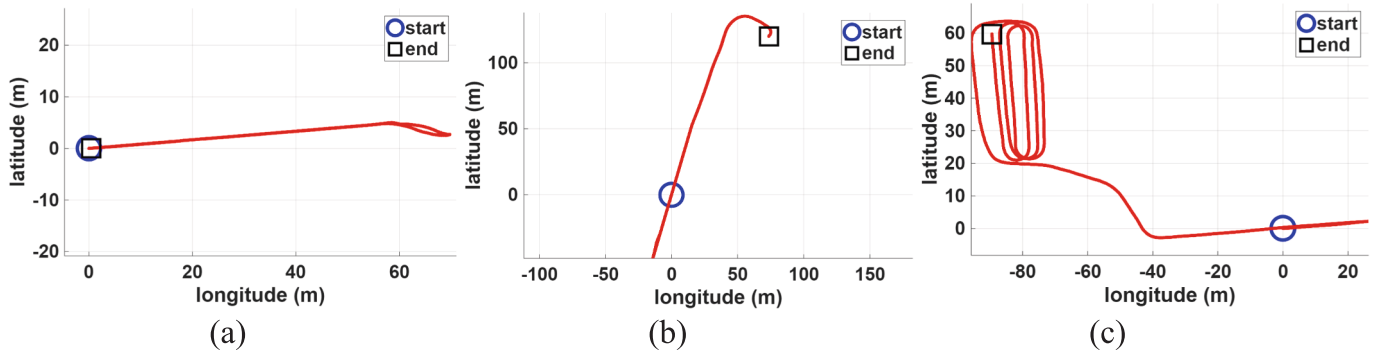


Fig. 14. The driving trajectories (a) T1, (b) T2, and (c) T3 for the test.

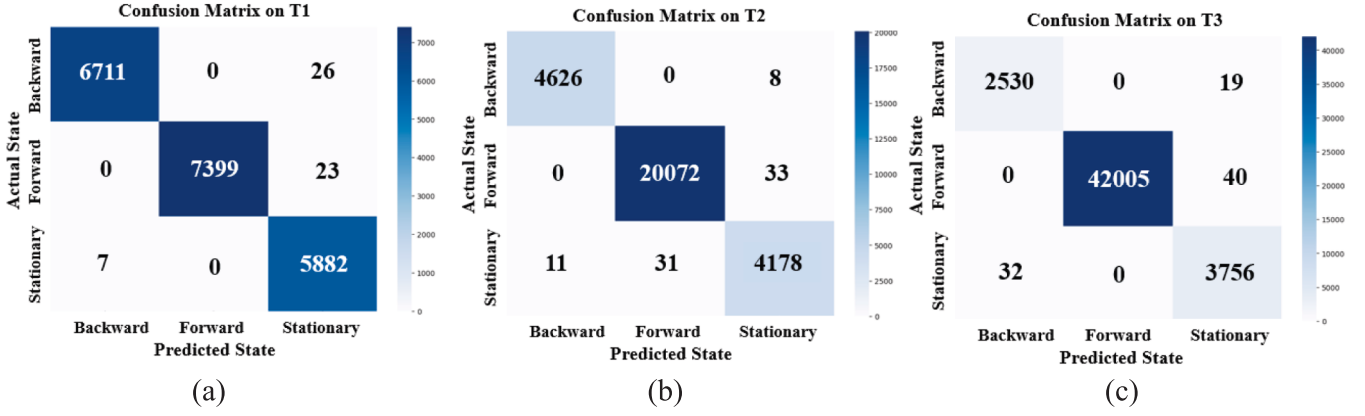


Fig. 15. Confusion matrix of driving state on (a) T1, (b) T2, and (c) T3.

Table 3

The accuracy of the proposed classification algorithm for trajectory T1, T2, and T3.

Trajectory	Driving state	Recall (%)	Precision (%)	F1-score (%)
T1 ($S^2 \rightarrow S^1 \rightarrow S^2 \rightarrow S^0 \rightarrow S^2$)	Backward	99.614	99.895	99.754
	Forward	99.690	100.000	99.844
	Stationary	99.881	99.173	99.526
	Average	99.728	99.689	99.708
T2 ($S^2 \rightarrow S^1 \rightarrow S^2 \rightarrow S^0 \rightarrow S^2$)	Backward	99.827	99.762	99.795
	Forward	99.835	99.845	99.840
	Stationary	99.004	99.028	99.016
	Average	99.555	99.545	99.550
T3 ($S^2 \rightarrow S^1 \rightarrow S^2 \rightarrow S^0 \rightarrow S^2$)	Backward	99.254	98.750	99.002
	Forward	99.904	100.000	99.952
	Stationary	99.155	98.453	98.803
	Average	99.438	99.068	99.252

the test datasets. S^0 , S^1 , and S^2 represent Forward, Backward, and Stationary, respectively. In addition, the driving trajectories T1, T2, and T3 for the test are shown in Fig. 14(a), (b), and (c), respectively.

4.2. Performance of the proposed classification algorithm

4.2.1. Performance analysis of the proposed algorithm

To train the proposed LSTM-based classification algorithm, acceleration and angular rate data were collected using an accelerometer and gyroscope while driving along 13 trajectories. The labeling of the train and test data was conducted based on the velocity measurements from the RTK-GPS. In this study, LSTM model is configured with 2 layers and a hidden state size of 20 and trained with an Adam optimizer, a learning

rate of 0.0001, a weight decay of 0.0001, a dropout rate of 0.1, and a sequence length of 10 for the data. The counts for Startback, Stop, and Startgo were set to 8, 18, and 8, respectively. The LSTM model was then tested using data from the other 3 trajectories: T1, T2, and T3 in Table 2. Fig. 15 shows the confusion matrix for each of the trajectory T1, T2, and T3, respectively. Additionally, Table 3 presents the Recall, Precision, and F1-score for trajectory T1, T2, and T3.

The LSTM model demonstrated a minimum accuracy of 98% and an average accuracy of over 99% for Recall, Precision, and F1-score for all trajectories. Specifically, for Recall, the highest accuracy is 99.904% for the Forward on trajectory T3, while the lowest was 99.004% for the Stationary on trajectory T2. For Precision, the Forward on trajectory T1 and T3 achieved the highest accuracy of 100%, whereas the Stationary on trajectory T3 had the lowest accuracy of 98.453%. Finally, for F1-score, the Forward on trajectory T3 reached the highest accuracy of 99.952%, while the Stationary had the lowest accuracy of 98.803%.

Table 4

The delay time of predicted driving states when driving state changes.

Trajectory	Driving state			
	Delay/early misprediction time (s)			
T1	Driving state	$S^2 \rightarrow S^1$	$S^1 \rightarrow S^2$	$S^2 \rightarrow S^0$
	Time duration (s)	+0.2582	+0.0660	+0.2040
T2	Driving state	$S^2 \rightarrow S^1$	$S^1 \rightarrow S^2$	$S^2 \rightarrow S^0$
	Time duration (s)	+0.0791	+0.1110	+0.3259
T3	Driving state	$S^2 \rightarrow S^1$	$S^1 \rightarrow S^2$	$S^2 \rightarrow S^0$
	Time duration (s)	+0.1940	+0.3201	+0.3680

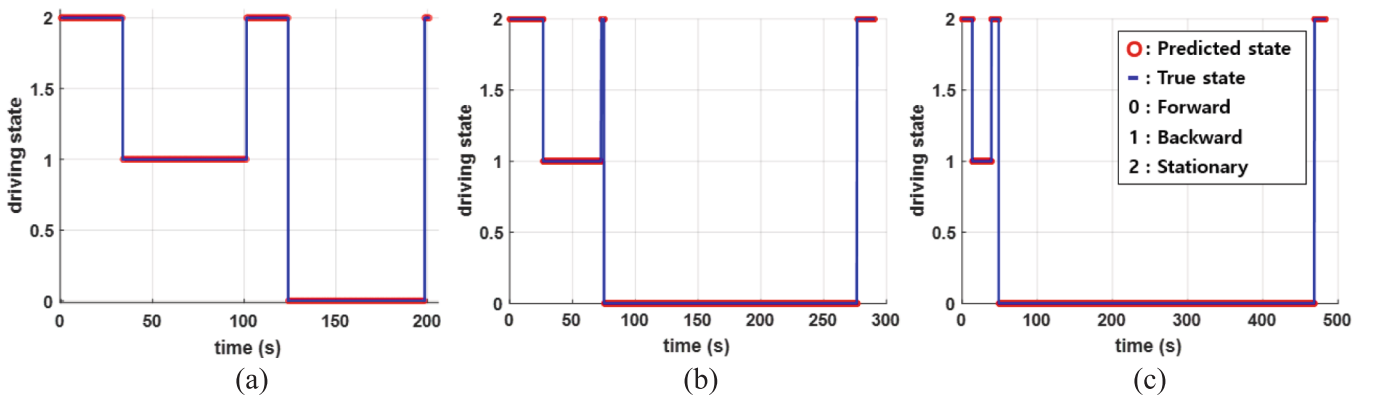


Fig. 16. Time-sequential comparison graph of the predicted driving state and the true driving state on (a) T1, (b) T2, and (c) T3.

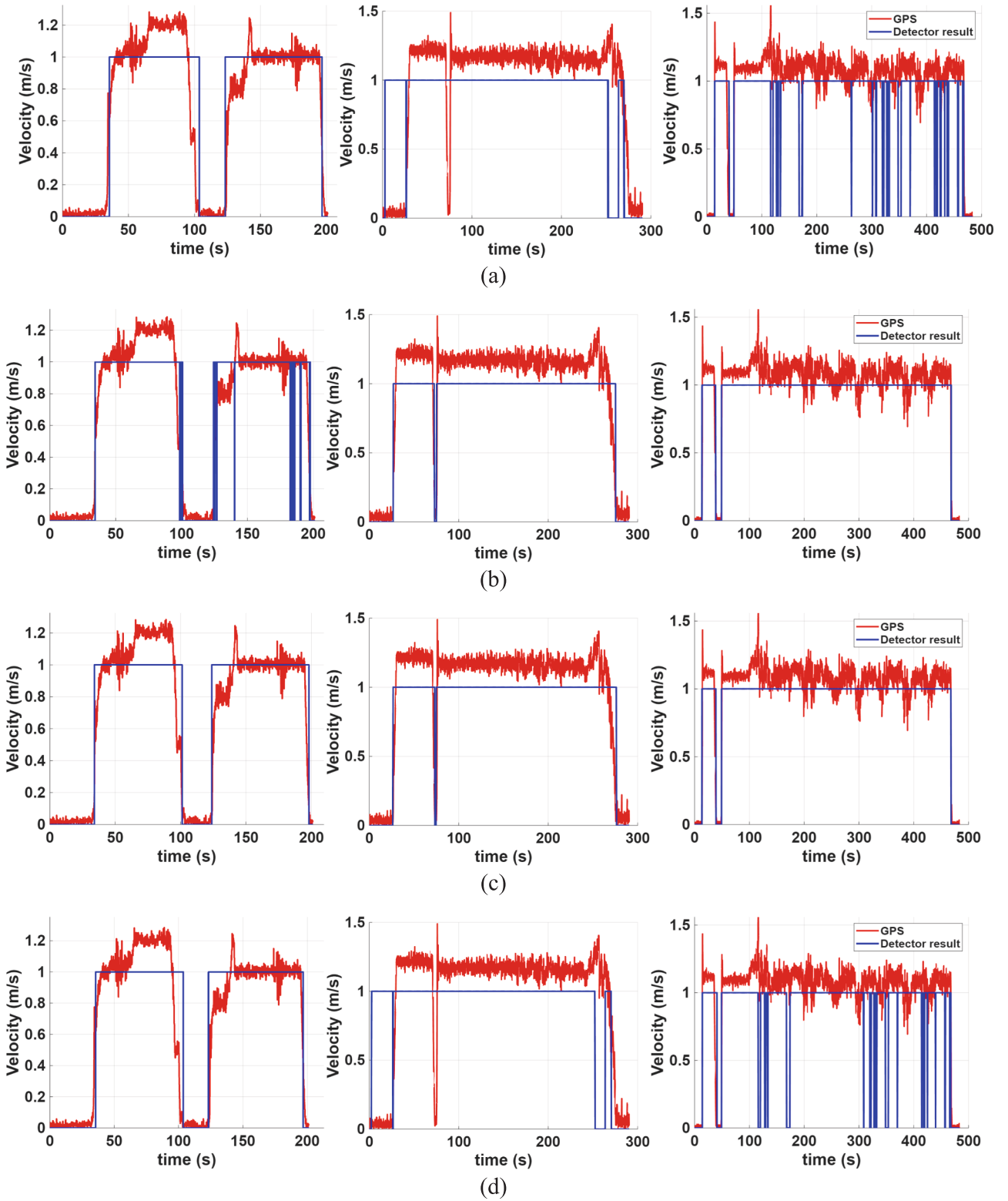


Fig. 17. Results of existing ZVDs for stationary detection. (a) Acceleration-Moving Variance Detector results, (b) Acceleration-Magnitude Detector results, (c) Angular Rate Energy Detector results, and (d) SHOE Detector results on T1, T2, and T3.

Fig. 16 shows the time-sequential graphs of the predicted and true driving states for trajectory T1, T2, and T3. In Fig. 16, the red dots represent the predicted driving state, while the blue line indicates the true driving state. The values 0, 1, and 2 on the y-axis correspond to the driving state of the tractor: Forward, Backward, and Stationary, respectively. As shown in Fig. 16, the proposed algorithm successfully

predicts the driving states of the tractor across all test trajectories.

In addition to accuracy, we also analyze each instance of misprediction. The algorithm proposed in this paper is designed to predict the real-time driving state of agricultural machinery, so it is essential to examine how quickly or slowly the predicted driving state aligns with the true driving state. Table 4 presents the delay/early misprediction

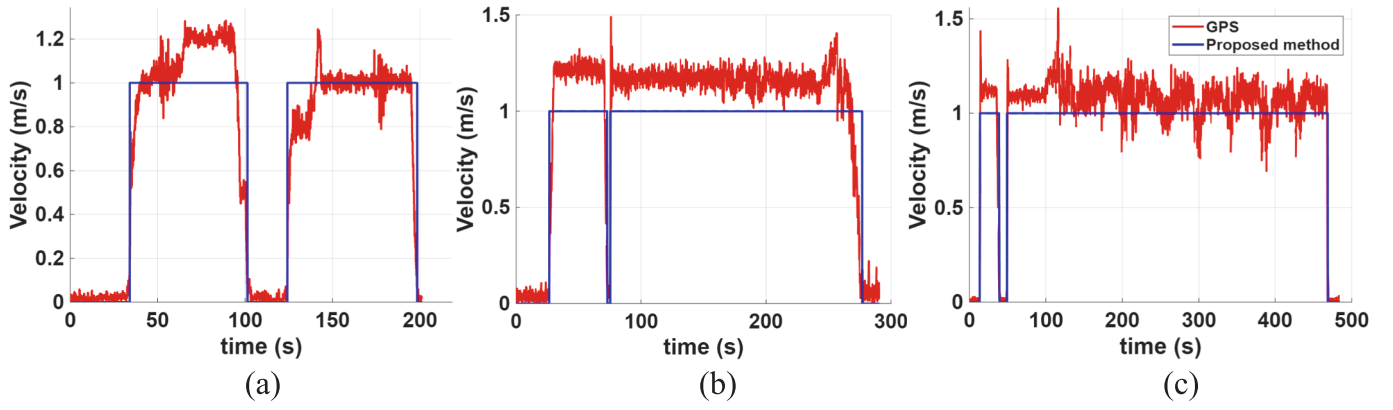


Fig. 18. Results of the proposed classification algorithm for stationary detection on (a) T1, (b) T2, and (c) T3.

Table 5

Delay time of the ZVD and proposed method with respect to RTK-GPS.

Detector	Trajectory	Driving state				Avg. time (s)
		Delay/early misprediction time (s)				
Acceleration-Moving Variance Detector	T1	$S^2 \rightarrow S^1$	$S^1 \rightarrow S^2$	$S^2 \rightarrow S^0$	$S^0 \rightarrow S^2$	1.6580
		+1.8581	+2.4310	−0.6340	−1.7090	
Acceleration-Magnitude Detector	T2	$S^2 \rightarrow S^1$	$S^1 \rightarrow S^2$	$S^2 \rightarrow S^0$	$S^0 \rightarrow S^2$	0.4423
		+0.2103	−0.1593	+0.2436	−1.1560	
Angular Rate Energy Detector	T3	$S^2 \rightarrow S^1$	$S^1 \rightarrow S^2$	$S^2 \rightarrow S^0$	$S^0 \rightarrow S^2$	0.3148
		+0.0799	−0.3541	+0.4954	−0.3300	
	T1	$S^2 \rightarrow S^1$	$S^1 \rightarrow S^2$	$S^2 \rightarrow S^0$	$S^0 \rightarrow S^2$	0.1886
		+0.3705	+0.0070	−0.0850	−0.2920	
T2	$S^2 \rightarrow S^1$	$S^1 \rightarrow S^2$	$S^2 \rightarrow S^0$	$S^0 \rightarrow S^2$	0.1649	
	−0.0283	+0.2130	−0.2514	−0.1670		
SHOE Detector	T3	$S^2 \rightarrow S^1$	$S^1 \rightarrow S^2$	$S^2 \rightarrow S^0$	$S^0 \rightarrow S^2$	0.2302
		+0.0459	+0.0767	+0.4155	−0.3830	
	T1	$S^2 \rightarrow S^1$	$S^1 \rightarrow S^2$	$S^2 \rightarrow S^0$	$S^0 \rightarrow S^2$	1.6600
		+1.8581	+2.3830	−0.6600	−1.7390	
Proposed method	T1	$S^2 \rightarrow S^1$	$S^1 \rightarrow S^2$	$S^2 \rightarrow S^0$	$S^0 \rightarrow S^2$	0.1395
		+0.2582	+0.0660	+0.2040	−0.0300	
	T2	$S^2 \rightarrow S^1$	$S^1 \rightarrow S^2$	$S^2 \rightarrow S^0$	$S^0 \rightarrow S^2$	0.2085
		+0.0791	+0.1110	+0.3259	+0.3180	
T3	$S^2 \rightarrow S^1$	$S^1 \rightarrow S^2$	$S^2 \rightarrow S^0$	$S^0 \rightarrow S^2$	0.2287	
	+0.1940	+0.3201	+0.3680	−0.0330		

Table 6

Information about the other ML models used for comparison.

ML only	SVM	SVM class S^0, S^1, S^2	RF class S^0, S^1, S^2
	RF		
ML with FSM	SVM with FSM	SVM class $\gamma^0, \gamma^1, \gamma^2, \gamma^3$	FSM class S^0, S^1, S^2
	RF with FSM	RF class $\gamma^0, \gamma^1, \gamma^2, \gamma^3$	FSM class S^0, S^1, S^2

time between the predicted driving states and the corresponding true driving state when the driving state changes. In Table 4, a “+” indicates a prediction that was made later than the true driving state, while a “−” indicates a prediction made earlier. The worst case of delay/early misprediction occurred in path T3 when the driving state changed from S^2 to S^0 , with a delay of +0.3680 s, and the shortest delay/early misprediction time occurred in path T1 when the driving state changed from S^0 to S^2 , with an early misprediction of −0.0300 s. The average Delay/early misprediction time across all trajectory is 0.1922 s. As shown in Fig. 16 and Table 4, the proposed classification algorithm demonstrates that the misprediction occur only during the transition phase of the driving state. Importantly, the duration of these mispredictions is

relatively short and the proposed algorithm can be applicable to real-time operation. We will further illustrate this through comparative studies with other methods as well as through navigation performance evaluation using actual tractor operations.

4.2.2. Performance comparison with conventional zero velocity detectors

The proposed classification algorithm distinguishes between the moving states (Forward, Backward) and stationary state (Stationary) of agricultural machinery, and therefore, it can also be expected to function as a ZVD. To verify the performance of the proposed algorithm, we conducted comparative studies with existing ZVDs [26,43,44]. Four types of ZVDs: Acceleration-Moving Variance Detector, Acceleration-Magnitude Detector, Angular Rate Energy Detector, and Stance Hypothesis Optimal Estimator (SHOE) Detector were considered in this study. First, the formula for the Acceleration-Moving Variance Detector is as follows:

$$T_v(z_n^a) = \frac{1}{L} \sum_{k \in \Omega_L} \|y_k^a - \bar{y}_n^a\|^2 < \gamma^v \quad (34)$$

$$\bar{y}_n^a = \frac{1}{L} \sum_{k \in \Omega_L} y_k^a \quad (35)$$

where y_k^a represents the acceleration data from the accelerometer at time

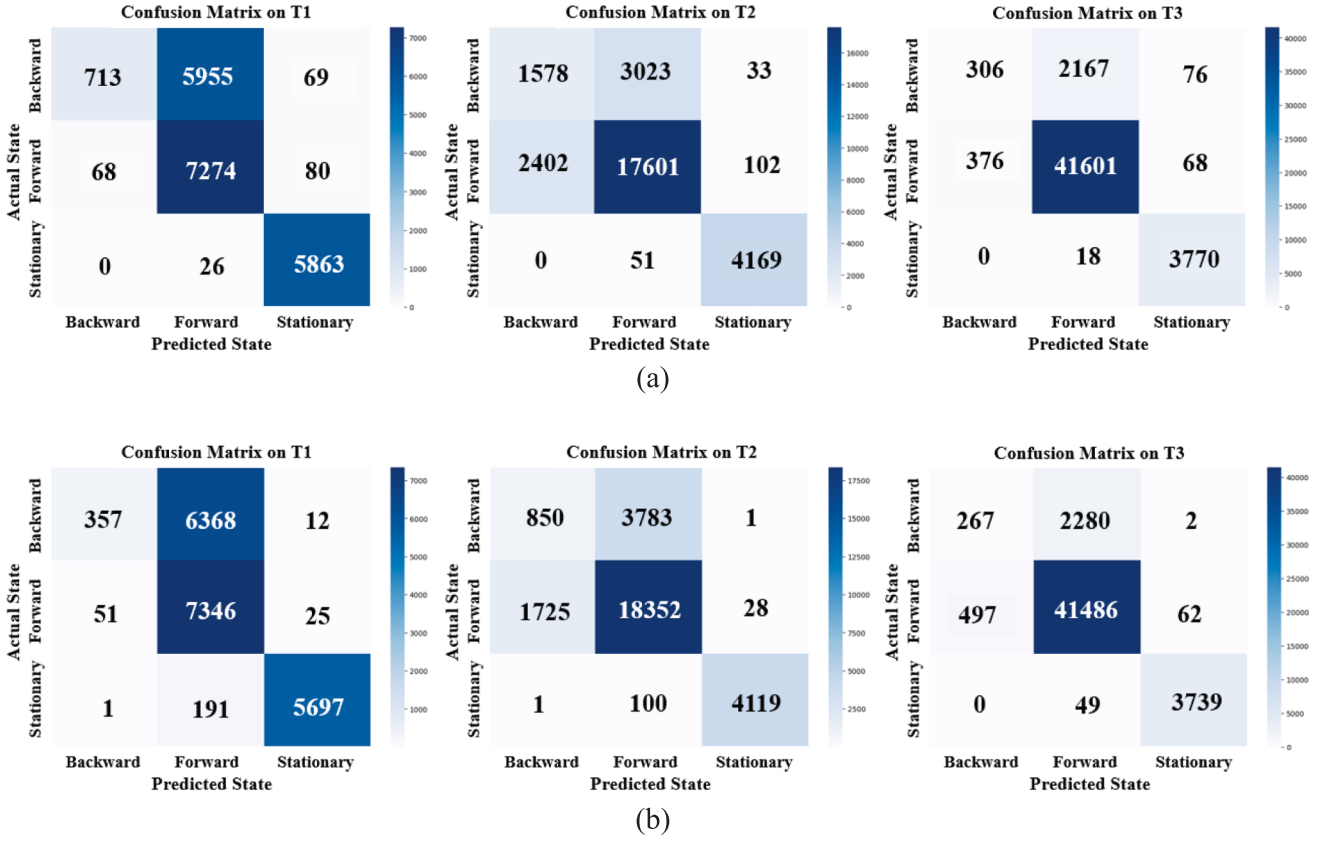


Fig. 19. Classification results for (a) SVM and (b) RF on T1, T2, and T3.

step k , L refers to the length of the sequence data, and γ^v denotes the threshold of Acceleration-Moving Variance Detector. If the inequality (34) is satisfied, it is determined that the object is currently stationary. Secondly, the Acceleration-Magnitude Detector is defined by the following equation:

$$T_m(z_n^a) = \frac{1}{\sigma_a^2 L} \sum_{k \in \Omega_L} (\|y_k^a\| - g)^2 < \gamma^m \quad (36)$$

where g represents gravity, σ_a^2 refers to the noise variance of the accelerometer, and γ^m denotes the threshold of Acceleration-Magnitude Detector. If the inequality (36) is satisfied, it is determined that the object is currently stationary. Thirdly, the Angular Rate Energy Detector is defined by the following equation:

$$T_w(z_n^w) = \frac{1}{L} \sum_{k \in \Omega_L} \|y_k^w\|^2 < \gamma^w \quad (37)$$

where y_k^w represents the angular velocity data from the gyroscope at time step k and γ^w denotes the threshold Angular Rate Energy Detector. If the inequality (37) is satisfied, it is determined that the object is currently stationary. Fourthly, the SHOE detector is defined by the following equation:

$$T(z_n) = \frac{1}{L} \sum_{k \in \Omega_L} \left(\frac{1}{\sigma_a^2} \|y_k^a - g \frac{\bar{y}_n^a}{\|\bar{y}_n^a\|}\|^2 + \frac{1}{\sigma_w^2} \|y_k^w\|^2 \right) < \gamma^s \quad (38)$$

where σ_w^2 represents the noise variance of the gyroscope, and γ^s denotes the threshold of SHOE Detector. If the inequality (38) is satisfied, it is determined that the object is currently stationary.

In this study, σ_a and σ_w are obtained from data measured by the IMU in a stationary state, with $\sigma_a = 0.0027 \text{ m/s}^2$ and $\sigma_w = 0.0002955 \text{ rad/s}$. The length of the sequence data and the threshold vary for each detector:

for the Acceleration Moving Variance Detector, $L = 260$ and $\gamma^v = 167.3$; for the Acceleration Magnitude Detector, $L = 80$ and $\gamma^m = 2061$; for the Angular Rate Energy Detector, $L = 70$ and $\gamma^w = 0.00006$; and for the SHOE Detector, $L = 255$ and $\gamma^s = 22,911,516$. For each detector, L , γ^v , γ^m , γ^w , and γ^s were selected to achieve the best performance, with N tested from 1 to 300. Figs. 17 and 18 present the results of the four ZVDs and the proposed algorithm. The blue line represents the results of the proposed algorithm and the ZVDs, where 0 indicates a stationary state, while 1 indicates a moving state. The velocity from the RTK-GPS is also plotted as a red line to indicate whether the tractor is moving or stationary. In these graphs, the results are shown in order from left to right for trajectory T1, T2, and T3. Among the four types of ZVD, the Angular Rate Energy Detector showed good performance across all three trajectories. In contrast, the remaining three ZVDs did not perform well on certain trajectory. The Acceleration Moving Variance Detector performed well on trajectory T1 but often confused stationary and moving states on trajectory T2 and T3. The Acceleration Magnitude Detector showed good performance on trajectory T2 and T3 but confused stationary and moving states on trajectory T1. The SHOE Detector only succeeded to distinguish between stationary and moving states on trajectory T1.

Table 5 shows a performance comparison of the stationary detection between the ZVDs and the proposed algorithm. The delay/early misprediction time is presented for trajectories where each detector successfully determined stationary and moving states. For the Acceleration Moving Variance Detector and SHOE Detector, L was relatively high, resulting in significantly longer average of delay and early misprediction time. Among the four detectors, the Angular Rate Energy Detector exhibited the best average delay and early misprediction time across all trajectories. A notable aspect of the result of the ZVDs is that, due to its reliance on a single predetermined threshold, it often predicts stationary or moving states in advance. If the stationary state is predicted earlier

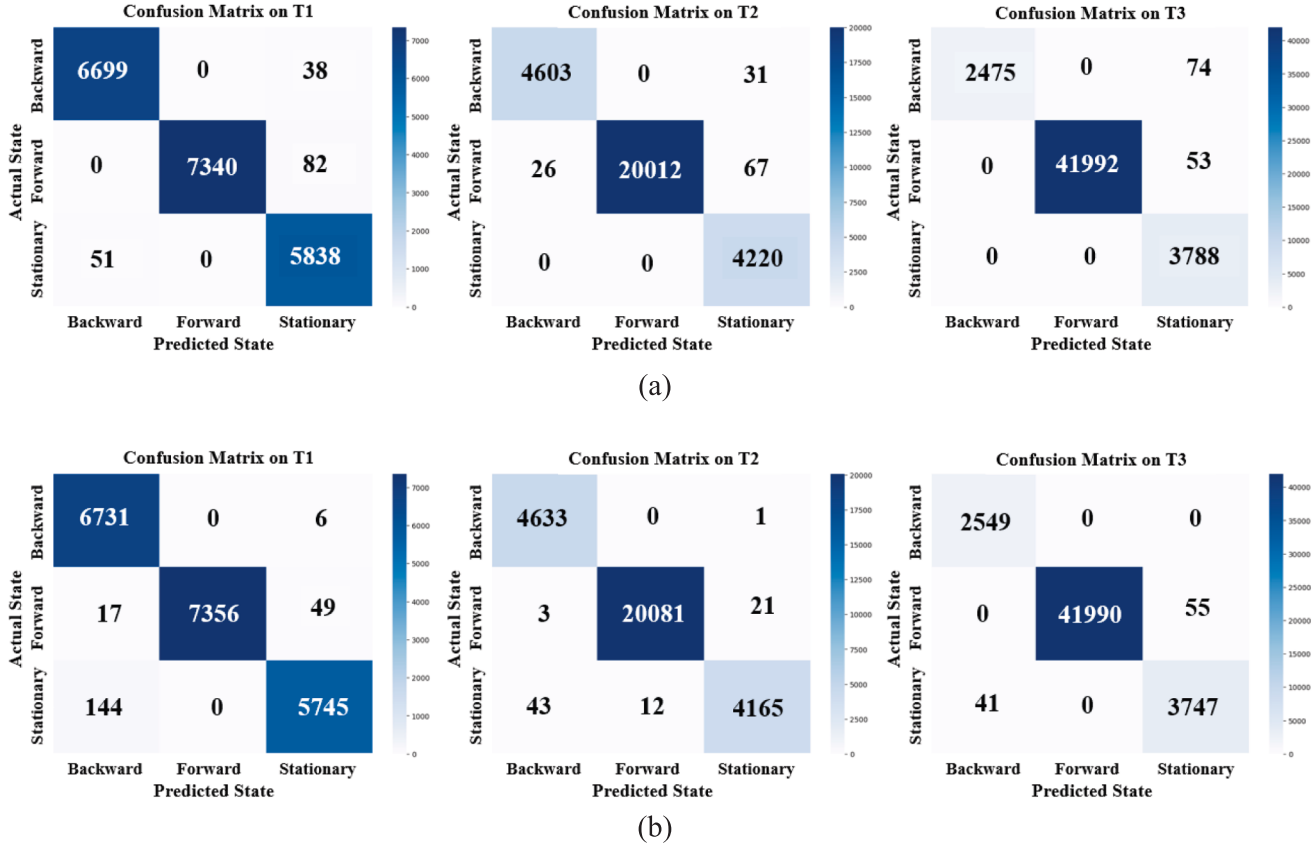


Fig. 20. Classification results for (a) SVM with FSM and (b) RF with FSM on T1, T2, and T3.

Table 7

Comparison of F1-score between other ML methods and proposed classification algorithm.

Algorithm	Trajectory	F1-score (%)		
		Backward	Forward	Stationary
SVM only	T1	18.967	70.358	98.529
	T2	36.638	86.321	97.817
	T3	18.941	96.937	97.896
SVM with FSM	T1	99.340	99.444	98.556
	T2	99.384	99.768	98.852
	T3	98.527	99.936	98.351
RF only	T1	9.991	68.889	98.029
	T2	23.578	86.688	98.446
	T3	16.118	96.636	98.511
RF with FSM	T1	98.774	99.553	98.297
	T2	99.495	99.910	99.084
	T3	99.202	99.934	98.735
Proposed method	T1	99.754	99.844	99.526
	T2	99.795	99.865	99.275
	T3	99.002	99.952	98.803

than the actual state and ZUPT is applied, it can lead to inaccurate position or attitude estimation of the agricultural machinery. On the other hand, the proposed method predicts earlier only for two cases ($S^0 \rightarrow S^2$ on T1 and $S^0 \rightarrow S^2$ on T3). Even in these two cases, the early misprediction time was only -0.03 s, which is much shorter than the early mispredictions from the ZVDs. When comparing the proposed method to the ZVDs, it was observed that the proposed method had shorter average delay/early misprediction time across all trajectories except for T2.

4.2.3. Performance comparison with machine learning-based methods

In this section, we demonstrate the superiority of the proposed

classification algorithm by comparing it with other machine learning-based methods. For comparative studies, two other machine learning-based methods: Support Vector Machine (SVM) and Random Forest (RF) are considered. SVM is a type of supervised learning that finds the optimal hyperplane to classify linear or nonlinear data. In this study, for SVM, a Radial Basis Function (RBF) Kernel is used to classify nonlinear data, with the regularization parameter set to 1.0, and the kernel coefficient (gamma) is automatically adjusted according to the variance of the data and the number of features. RF is a machine learning method that constructs multiple decision trees to solve classification and regression problems. For RF, the model is composed of 40 decision trees, and the random state is set to 42 to ensure consistent results with each training iteration. For both models, the sequence length of the data was set to 10, and the parameters were determined through iterative adjustments to achieve best performance for each model. Furthermore, we consider two other models where SVM and RF are combined with FSM, respectively. Table 6 provides a summary of the machine learning (ML) models used for comparison with the proposed classification algorithm and the classes used in each model.

As shown in Table 6, we compared a total of 4 models with the proposed classification algorithm. S^0 , S^1 , and S^2 in Table 6 represent Forward, Backward, and Stationary, respectively. When the ML models combined with FSM, as in the proposed classification algorithm, SVM and RF first predict the maneuver state of the agricultural machinery, respectively. The FSM then utilizes these predictions to determine the driving state of the agricultural machinery. Here, Y^0 , Y^1 , Y^2 , and Y^3 represent Startgo, Startback, Stop, and Move, respectively. Like the proposed classification algorithm, all 4 comparison models used acceleration and angular rate data as input $x = [y_x^a \ y_y^a \ y_z^a \ y_x^w \ y_y^w \ y_z^w]^T$, and they utilized sequence data of length N . Fig. 19 shows the results of classifying the driving states of the tractor along trajectory T1, T2, and T3 using ML models, where Fig. 19(a) represents the results for SVM,

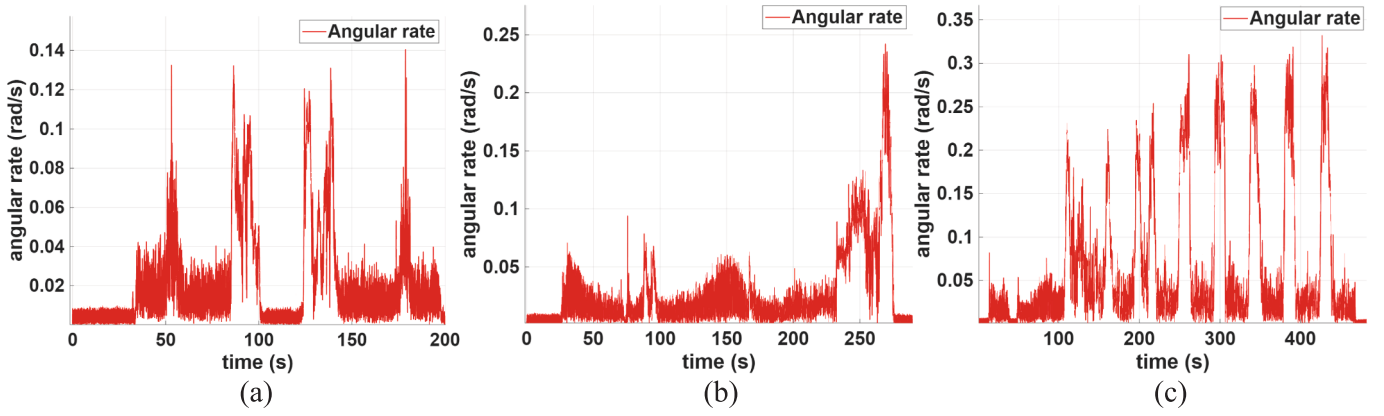


Fig. 21. Angular rate magnitude for trajectories (a) T1, (b) T2, and (c) T3.

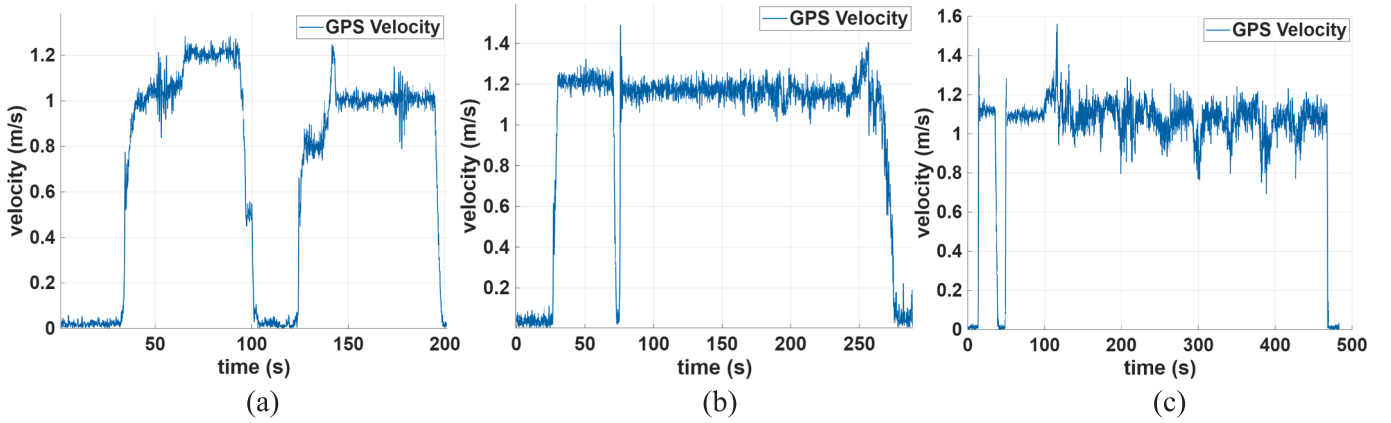


Fig. 22. GPS velocity of trajectories (a) T1, (b) T2, and (c) T3.

Fig. 19(b) represents the results for RF. Fig. 20 shows the results of classifying the driving states of the tractor along trajectory T1, T2, and T3 when using ML models with FSM, where Fig. 20(a) represents the results for SVM with FSM, Fig. 20(b) represents the results for RF with FSM. Additionally, Table 7 compares the F1-scores of the predicted driving states from the four models with those obtained from the proposed classification algorithm.

The SVM and RF models without using FSM achieve F1-scores above 97% for Stationary detection. However, these models perform poorly in distinguishing between the Backward and Forward driving states. When examining the prediction results of the models that incorporate FSM, both SVM with FSM and RF with FSM show improved performance compared to the models without using FSM, as shown in Fig. 20 and Table 7. Among the ML-based methods, the proposed algorithm demonstrates the best results for driving state prediction, except for the Backward driving state prediction of T3 and the Forward driving state prediction of T2.

4.2.4. Discussion

In this section, the performance of the proposed classification algorithm is comprehensively discussed through comparisons with conventional ZVDs and other machine learning-based methods. Overall, the proposed method reliably distinguishes the stationary, forward, and backward states. Unlike conventional ZVDs, which can only distinguish between stationary and moving states, the proposed method provides extended functionality by additionally distinguishing forward and backward motion. Furthermore, when compared with other machine learning-based methods, the proposed classification algorithm generally showed higher classification performance.

A comparison with conventional ZVDs shows that the proposed method yields smaller average delay and fewer early mispredictions on all trajectories except T2. As shown in Table 5, the angular rate energy detector showed slightly better performance than the proposed method on trajectory T2 in terms of average delay/early misprediction time. Among the evaluated trajectories, T2 corresponds to a relatively simple motion profile, characterized by low variations in both vehicle speed and angular rate within each driving state as shown in Figs. 21 and 22. In such simple motion scenarios, threshold-based methods are particularly effective because the signal patterns are stable and easily separable using fixed thresholds.

However, for trajectories with more complex and dynamic motion, the proposed LSTM-based method consistently outperforms the conventional threshold-based ZVDs due to its ability to capture temporal patterns and transitions in IMU signals more effectively. Notably, although the angular rate energy detector achieves the best accuracy for T2 in terms of the average time difference between the true driving state transition and the detected transition, it exhibits larger early misprediction compared to the proposed method. Such premature detection can trigger early ZUPT application, which may degrade position and attitude estimation accuracy in the agricultural machinery.

Furthermore, as shown in Table 7, the proposed method achieves the highest F1-score in most cases. While RF with FSM slightly outperforms the proposed method for the Forward motion of T2 and the Backward motion of T3 by 0.045% and 0.200%, respectively, these differences are negligible. Therefore, these cases can be regarded as minor, trajectory-specific variations rather than deviations from the overall performance trend of the proposed method.

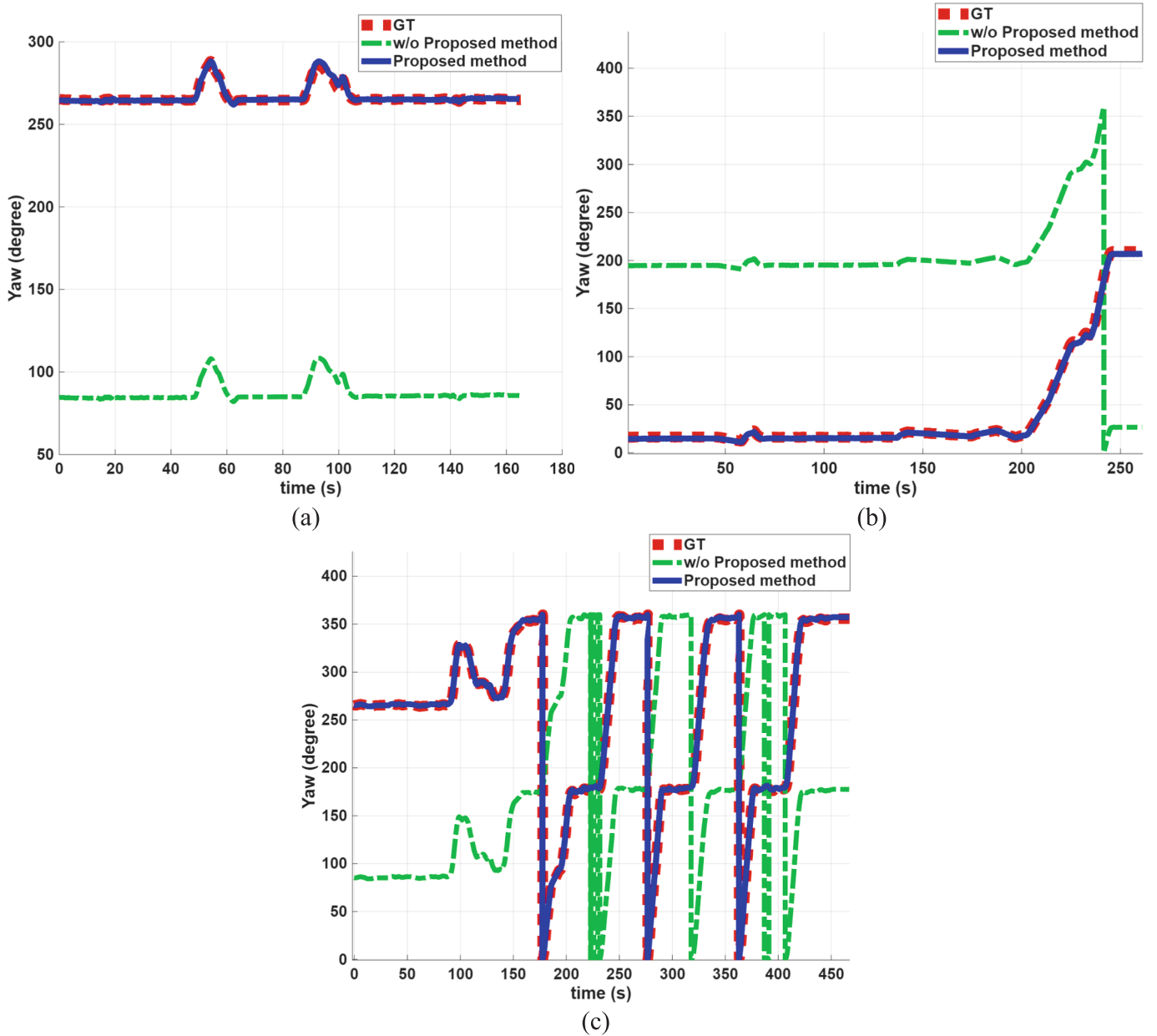


Fig. 23. Results of yaw angle estimation with and without the proposed classification algorithm on (a) T1, (b) T2, and (c) T3.

Table 8

RMSE of the estimated position and attitude with the proposed method.

Trajectory	Position RMSE (m)	Yaw angle RMSE (°)	Roll angle RMSE (°)	Pitch angle RMSE (°)
T1	0.0256	0.4500	0.3219	0.2437
T2	0.0231	1.1126	0.6031	0.3694
T3	0.0135	1.2511	0.2876	0.4567

4.3. Performance of the proposed navigation system

This section evaluates the performance of the INS/GPS integrated navigation system with the proposed driving state classification algorithm. In the test trajectories T1, T2, and T3, the tractor always starts with backward motion and includes a stationary interval during operation. The performance of the proposed navigation system over these three trajectories is evaluated in two aspects. First, we assess the initial yaw alignment performance when the tractor starts moving in backward

motion. Second, we compare the position estimation performance under a GPS outage.

4.3.1. Performance of yaw angle estimation

We compare the yaw estimation performance with and without the proposed algorithm and examine the position and attitude estimates when the proposed driving state classification algorithm is applied. For tractors equipped with auto-steering but without access to in-vehicle data, the initial yaw alignment is performed using only GPS measurements. As a result, when the vehicle starts moving in a backward state, the system fails to achieve correct initial yaw alignment. In contrast, the system with the proposed classification algorithm can identify backward motion in real time, enabling accurate initial yaw alignment even when the tractor starts moving backward. These results are illustrated in Fig. 23, which compares the yaw angle estimation results with and without the proposed method against the ground truth for trajectories T1, T2, and T3. The red line represents the GT yaw angle, the green line denotes the yaw angle estimated without the proposed method, and the

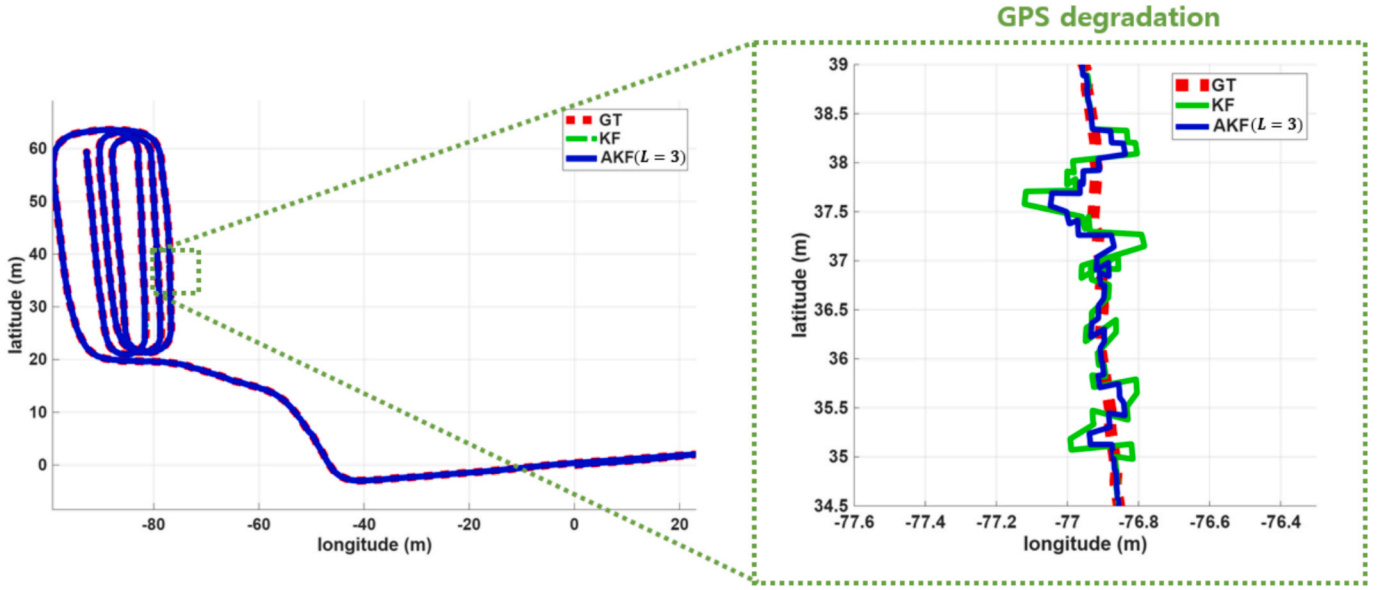


Fig. 24. Position estimates under GPS degradation with the KF and the proposed AKF.

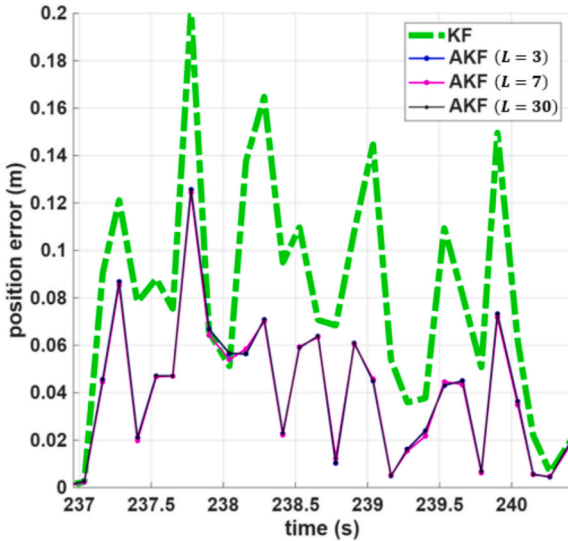


Fig. 25. Position estimation errors under GPS degradation with the KF and the proposed AKF.

blue line corresponds to the yaw angle estimated with the proposed method. Table 8 summarizes the RMSE values for estimated position and attitude when the proposed method is applied. The position and yaw errors are within 0.03 m and 1.3° , respectively, and the roll and pitch errors are within 1° .

4.3.2. Performance comparison under GPS degradation and outage conditions

The performance of the proposed adaptive Kalman filter under GPS degradation conditions was further analyzed, along with the effect of the sliding window length L . We evaluated the proposed AKF with different values of L and compared its performance with that of a conventional KF using a constant measurement noise covariance. To simulate GPS degradation, random Gaussian noise with zero mean and a standard deviation of 0.1 m was artificially added to the longitude and latitude measurements of the GPS over the 3 s interval from 237 s to 240 s in trajectory T3. It is noted that a regularization term ϵ was added to the measurement noise covariance adaptation Eq. (19) to prevent numerical

instability [45].

Figs. 24 and 25 show the position estimation results and the corresponding position estimation errors of the KF and the proposed AKF under GPS degradation conditions. The KF shows limited robustness to temporary GPS degradation, whereas the proposed AKF demonstrates more stable and robust estimation performance. Quantitatively, the mean and maximum position errors of the KF are 0.0913 m and 0.2007 m, respectively, while those of the proposed AKF are reduced to 0.0459 m and 0.1257 m, respectively.

Furthermore, Fig. 25 illustrates the effect of the sliding window length L on estimation error. When the sliding window length L is excessively small, the numerical stability of the filter deteriorated. In particular, divergence occurs when $L \leq 2$ in the evaluated trajectories. For $L \geq 3$, the performance differences become marginal. Therefore, in this study, $L = 3$ is selected by considering both estimation accuracy and computational efficiency.

Next, we compare the position estimation results with and without the proposed classification algorithm under a GPS outage. A conventional navigation system of auto-steering system identifies stationary intervals only from GPS measurements and applies corrections such as zero velocity update based on this information. Therefore, when GPS signals weaken or drop out, these corrections cannot be carried out. As a result, estimation errors accumulate throughout the outage, ultimately degrading the navigation performance. In contrast, with the proposed driving state classification algorithm, stationary intervals can be detected independently of GPS availability, which enables timely zero velocity and zero angular-rate updates (ZVZARU) and suppresses error growth. To verify this, we imposed an artificial GPS outage of about 4 s on a segment of T3, consisting of roughly 1.5 s of backward motion followed by about 2.5 s of stationary state. In the navigation system without driving state classification, a stationary interval is declared when the GPS velocity magnitude falls below a predefined threshold, which is set to 0.07 m/s. Figs. 26 and 27 present the position estimates and the corresponding position errors with and without the classification algorithm during the GPS outage. In the navigation system without driving state classification, the stationary state is not detected during the GPS outage and consequently IMU bias and noise cause continuous error growth. In contrast, when the classification algorithm is applied, the navigation system detects the stationary state even without GPS and performs ZVZARU, effectively suppressing the drift of the position error. As a result, during the stationary interval, the estimated trajectory remains close to the reference trajectory and the error curve becomes

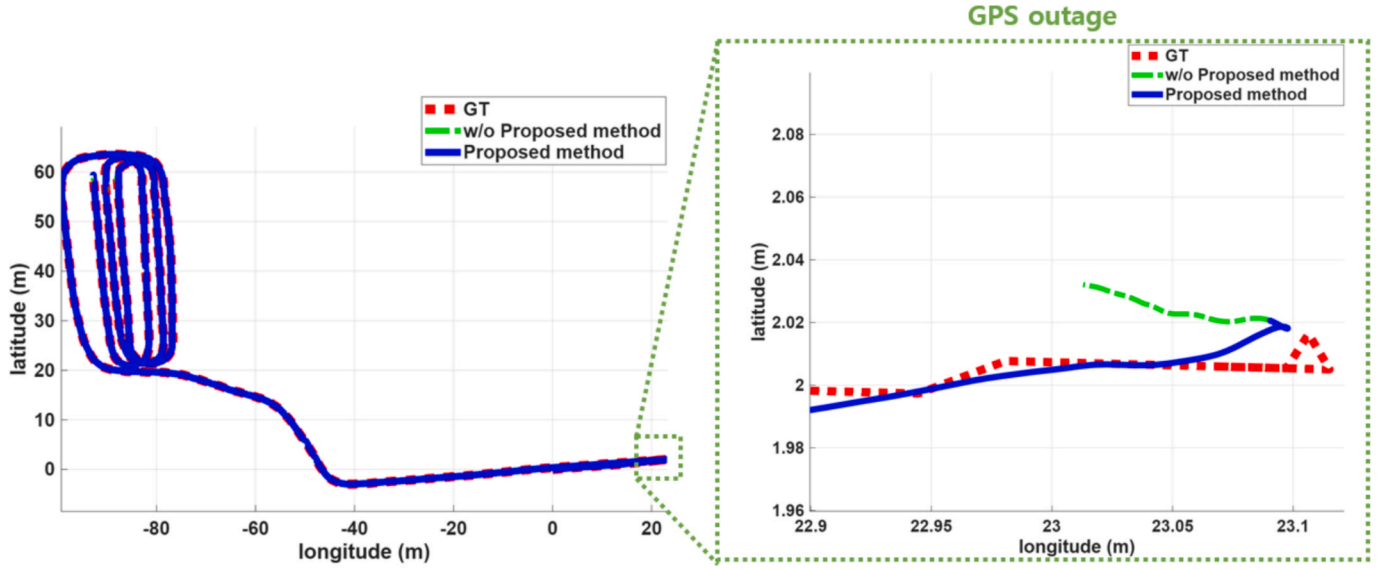


Fig. 26. Position estimates under a GPS outage with and without the proposed algorithm.

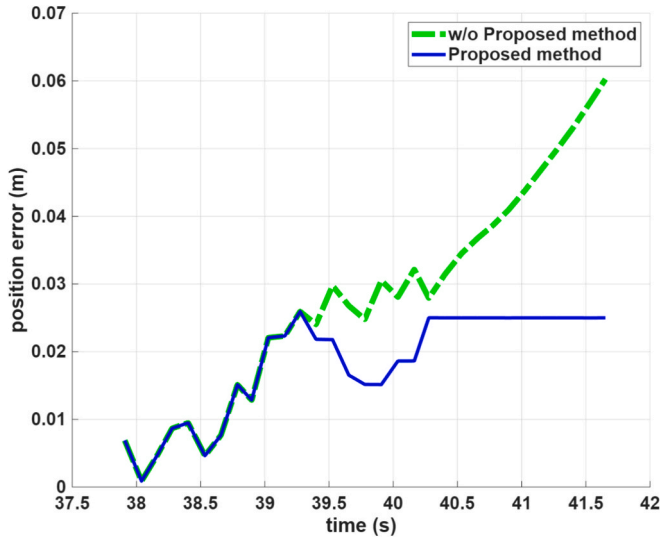


Fig. 27. Position estimation error under a GPS outage with and without the proposed algorithm.

Table 9

Mean and maximum position errors over the GPS outage in trajectory T3.

Method	RMSE (m)	Max error (m)
Proposed method	0.0184	0.0259
w/o Proposed method	0.0277	0.0603

noticeably flatter. Quantitatively, the position RMSE decreases from 0.0277 m to 0.0184 m, a reduction of 33.5%, and the maximum position error decreases from 0.0603 m to 0.0259 m, a reduction of 57.0%. Table 9 summarizes the RMSE and maximum errors over the outage interval.

5. Conclusion

In this paper, we proposed an IMU-based LSTM-FSM driving state classification algorithm that identifies stationary, forward, and backward motions of a semi-autonomous tractor in real time. The algorithm

was trained and evaluated using real tractor driving data collected from farm roads and farm fields, and the test results showed very high performance, with F1-scores ranging from 98.803% to 99.952%. To verify the performance and robustness of the proposed algorithm, comparative studies were conducted against conventional ZVD methods and other machine learning-based methods. Among the conventional ZVD methods, the acceleration moving variance detector, acceleration magnitude detector, and SHOE detector frequently failed to detect stationary intervals and thus exhibited poor performance. Even when compared with the angular rate energy detector, which showed relatively good performance, the proposed algorithm outperformed it on all trajectories except for one. In addition, the proposed algorithm can accurately distinguish forward and backward driving, which conventional ZVD methods cannot classify, providing a clear advantage. For the machine learning-based methods, the models alone showed low performance in distinguishing forward and backward motions. Although their performance improved when combined with the proposed FSM, their average accuracy still remained lower than that of the proposed LSTM-FSM driving state classification algorithm on most trajectories.

The proposed classification algorithm was integrated into an adaptive Kalman filter framework to evaluate the overall navigation performance. Under normal driving conditions, the RMSE of position and attitude estimation with respect to the ground truth data remained within 0.03 m and 1.3°, respectively. Moreover, unlike GPS-based commercial navigation systems, the proposed system successfully performed initial alignment not only during forward motion but also during backward motion. The performance of the proposed algorithm was evaluated under GPS degradation and outage conditions. The results showed that, under GPS degradation conditions, the proposed adaptive Kalman filter responded more robustly than the Kalman filter using a constant measurement noise covariance, significantly reducing the mean and maximum position errors. Furthermore, during GPS outage conditions, the proposed classification algorithm successfully detected stationary intervals, thereby enabling the application of the correction algorithm based on ZVZARU. As a result, position estimation accuracy was significantly improved during GPS outages, effectively suppressing error growth. These experimental results clearly demonstrate that the proposed algorithm has practical applicability and robustness in real agricultural environments.

Future research will focus on extending the algorithm to classify more complex driving states, such as left and right turns, rapid acceleration and deceleration, and more dynamic or aggressive maneuvers. In

addition, control algorithms based on the developed navigation system will be investigated to further improve the precision of autonomous driving in agricultural machinery.

CRedit authorship contribution statement

Euijun Jung: Writing – original draft, Software, Methodology, Investigation, Data curation. **Yongjin Kwon:** Software, Methodology. **Wonseok Choi:** Software, Investigation. **Minjae Kwak:** Software, Investigation. **Chang-Ho Lee:** Validation, Investigation. **Woongsun Jeon:** Writing – review & editing, Writing – original draft, Supervision, Software, Methodology, Funding acquisition, Formal analysis, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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